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The Greenness-of-Jobs Index (GOJI): A Task-Based Approach to Measure the Green Transition Exposure of Occupations and the Greening of The Labour Market

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26 June 2026

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Abstract

Global efforts toward a low-carbon and greener economy are reshaping work by altering the task content of occupations. This paper introduces the *Greenness-of-Jobs Index (GOJI)*, a task-based indicator that quantifies the share of environment-related tasks in each occupation in Germany. Using task information from the BERUFENET database, I apply natural language processing to classify tasks as green, brown, or white based on international standards and to calculate their occupational-level shares. I trace GOJI’s evolution—from its initial version in 2018 (GOJI 1.0) through GOJI 2.0 and 3.0 to the current GOJI 4.0, available annually from 2012 to 2025. Using concrete examples, I demonstrate the classification process and coding logic. I situate GOJI within the literature on green jobs, skills, and tasks, and highlight its methodological advances over output-based and previous task-based indices. GOJI’s task-level design supports strategic foresight into the technological and social transitions underpinning sustainable labor market transformation.

Keywords: Green jobs; Task-based index; Occupational tasks; Environmental sustainability; Labor market transition; Natural language processing

1 Introduction

Climate change and environmental degradation have triggered widespread efforts toward greener economies and sustainable development. As industries implement cleaner technologies and more resource-efficient processes, the nature of work is changing. Jobs across diverse sectors are evolving to include “green” tasks oriented toward environmental protection, energy efficiency, and pollution reduction—or to leverage the potential of existing green tasks. International institutions such as the [International Labour Organization](#) (2018) and the [Organisation for Economic Co-operation and Development](#) (2023) emphasize that these developments have profound implications for the world of work. The need to measure the environmental content of jobs is therefore pressing, as highlighted by frameworks developed by the [United Nations Environment Programme](#) (2011). These transformations raise key questions for labor markets: To what extent are occupations becoming greener, which remain entrenched in environmentally harmful activities, and what does this mean for employment, wages, and the demands for vocational and continuing education? Addressing these questions requires robust, scalable, and policy-relevant measurement approaches. Reliable metrics are essential to anticipate workforce needs, monitor transition pathways, and guide vocational and innovation policy.

Defining and measuring “green jobs” has proven challenging. Early efforts often focused on sectors or outputs, labeling entire jobs or industries as green based on the production of environmental goods and services. For example, the U.S. Bureau of Labor Statistics (BLS) distinguished between jobs producing green outputs (e.g., renewable energy equipment) and jobs involving green processes (e.g., roles that improve a firm’s environmental performance) ([U.S. Bureau of Labor Statistics, 2013](#)). International bodies like the International Labour Organization (ILO) offered broader definitions, adding that truly *green jobs* should also meet standards of decent work ([International Labour Organization, 2018](#)). However, it has proven difficult to measure both aspects together, and this approach may have impeded efforts to quantify green jobs. All output-based approaches struggle to capture the *greening of occupations* within traditional sectors. Many conventional jobs are gradually incorporating environmentally friendly practices, even if

their primary output is not explicitly green but rather environmentally neutral or even “brown.” Process-oriented approaches were introduced to account for this integrated environmental activity (Consoli et al., 2016). Yet, existing metrics still often treat jobs as monolithically green or not, without capturing gradations of environmental content.

This paper advances a task-based approach to measuring the greenness of jobs. Rather than classifying entire occupations or industries as green, I focus on the composition of tasks within each occupation. Tasks are the unit of work activities that produce outputs, and they can change over time even if an occupation’s title remains the same. I differentiate tasks that are explicitly environmental (“green tasks”) from those that are environmentally detrimental (“brown tasks”) or neither (“white tasks”). By quantifying the share of green and brown tasks in each occupation, I construct a continuous **Greenness-of-Jobs Index (GOJI)** for every occupation and track how it evolves. This task-based index provides a more granular and dynamic measure of occupational greenness than prior binary labels. Analogous to the skills taxonomies used by the European Commission and the OECD, the GOJI differentiates between “green tasks,” which are beneficial to the environment or climate, “white tasks,” which are considered neither environmentally beneficial nor harmful, and “brown tasks,” which harm the environment and/or global climate (European Commission, 2022, Organisation for Economic Co-operation and Development, 2023). This typology provides a conceptual foundation for assessing occupational roles in the green transition of labor markets.

This paper documents **GOJI 4.0** (current version), which covers German occupations annually from 2012 to 2025. GOJI was first published in 2018 as a green-only index (Janser, 2018, 2019) and has since undergone three successive refinements; the full methodological history is documented in Appendix B.

Prior task-based approaches — including Dierdorff et al. (2009) and Peters (2014) for the United States and Vona et al. (2018) using O*NET — establish that green occupations are generally knowledge-intensive and not categorically distinct from comparable non-green occupations in their skill requirements. Consoli et al. (2016) further find that green jobs demand above-average cognitive and interpersonal skills. GOJI extends

this tradition to the German labour market, exploiting the BERUFENET occupational database, which provides detailed, annually updated task descriptions not available elsewhere. Unlike O*NET-based measures, GOJI uses the official BA Green Skills list for green-task identification and a curated brown dictionary for brown tasks, making classification directly traceable to authoritative sources.

My findings reveal substantial heterogeneity in occupational greenness, with a sharp rise in green-classified tasks following the Federal Employment Agency’s green-leaf roll-out in the 2023–2024 BERUFENET snapshots. The majority of German occupations remain predominantly white (no classified tasks), while a minority of occupations in renewable energy, construction, and environmental services accumulate substantial green task shares, and a smaller set in fossil-fuel extraction, heavy industry, and livestock production carry significant brown task shares.

The remainder of this report is organised as follows. Section 2 defines green, brown, and white tasks and situates the approach in the context of international measurement frameworks. Section 3 describes the data source (BERUFENET), the classification procedure, and the index formula, with illustrative examples from real occupations. Sections 4–6 present aggregated results: by occupational segment and requirements level (Section 4), by WZ08 one-digit industry sector (Section 5), and by German federal state (Section 6). Section 7 concludes. Appendices cover example task classifications (App. A), the methodological history across all four GOJI versions (App. B), additional descriptors (App. C), availability in the IAB Occupational Panel (App. D), a side-by-side version comparison (App. E), and a review of related literature on green jobs and occupational greenness (App. F).

2 Conceptual Foundations: Green, Brown, and White Tasks

2.1 Defining Green, Brown and White Tasks

A central premise of this approach is that jobs can be understood as bundles of tasks, some of which have an environmental orientation and others do not.

Green tasks involve the application of occupational skills, practices, and knowledge with the potential to protect or improve the environment and/or climate, or to help adapt to climate change in an eco-friendly way. They can be found, for example, in the context of renewable energy and renewable resources, energy efficiency, material efficiency, recycling, and nature and species conservation. All of the green tasks used for the GOJI are classified by expert sources as being environmentally and/or climate-friendly and/or necessary for sustainable climate adaptation. Real examples of classified green tasks from the 2025 BERUFENET snapshot include: *Fotovoltaikanlagen* (photovoltaic systems; renewable energy), *Ökologischer Landbau* (organic farming; renewable resources), *Energetische Gebäudedämmung* (thermal insulation of buildings; energy efficiency), *Material- und Ressourceneffizienz-Konzepte entwickeln* (developing material and resource efficiency concepts; material efficiency), *Recycling* (recycling), and *Forstschutz* (forest protection; nature and species conservation).

Brown tasks involve occupational skills, practices, and knowledge that have the potential to harm the environment and/or climate. Brown tasks can be found, for example, in the context of fossil fuel combustion technologies, energy-intensive processes, land sealing, and the use of environmentally harmful substances. The brown tasks used for the GOJI are classified as environmentally or climate-harmful by expert sources. Real examples of classified brown tasks from the 2025 BERUFENET snapshot include: *Kohleveredelung* (coal upgrading; fossil fuel combustion technologies), *Zement herstellen* (cement production; energy-intensive processes), *Asphaltieren* (asphalt laying; land sealing), and *Gerben* (leather tanning using chromium salts; environmentally harmful substances).

Many tasks in the economy are **white**, meaning they do not have an obvious positive

or negative environmental dimension. These might be general administrative, managerial, or technical tasks that could be present in any job regardless of industry (e.g. *managing a team, preparing financial accounts, or training new staff*). White tasks neither advance environmental goals nor significantly worsen environmental outcomes. In my framework, white tasks form the baseline of an occupation’s task set, against which the presence of green or brown tasks is measured.

By classifying tasks into these three categories, I acknowledge that greenness is not an all-or-nothing property of a job, but a matter of degree. An occupation might have a mix of task types. For instance, consider a building electrician: if they spend part of their time installing energy-efficient LED lighting or solar panels (green tasks) and part of their time wiring conventional electrical systems (white tasks), that occupation has a certain green task share. On the other hand, a coal plant maintenance technician might perform equipment repair (white task) but also handle coal combustion systems (brown tasks), giving that occupation a brown task component. The concept of “greening of jobs” refers precisely to changes in this mix—if the electrician’s role evolves to focus more on renewable energy systems over time, its green share rises; if the coal technician’s role shifts towards managing emissions control equipment (slightly greener tasks) or if the occupation itself declines, the labor market is said to green.

It is worth noting that my use of the terms *green* and *brown* tasks parallels how some studies describe “clean” versus “dirty” industries or jobs. However, I apply it at the granular task level. This is conceptually powerful because it allows virtually any occupation to have a greenness dimension—unlike earlier definitions of *green jobs* that might categorically exclude, say, accountants or drivers, unless those jobs are in a renewable energy company. Under a task-based approach, even an accountant’s job could have a green component (if, for example, they specialize in carbon accounting or sustainability reporting, those tasks would be green) or a brown component (though in an office job this is less likely unless indirectly via the sector’s nature). In most cases, occupations like accountants would mainly have white tasks.

2.2 Key Concepts: Green Transition Exposure and Greening of Occupations

Green transition exposure refers to the extent to which an occupation is affected—directly or indirectly—by the societal shift toward an environmentally sustainable economy. This exposure may stem from regulatory changes, the adoption of green technologies, the phasing out of carbon-intensive activities, or shifts in consumer preferences and production processes. Importantly, exposure can be positive (e.g., increased demand for renewable energy technicians) or negative (e.g., declining relevance of coal-related tasks). In the context of GOJI, green transition exposure is captured by the presence and weight of *green* and *brown* tasks in an occupation’s profile, indicating both its potential for transformation and the direction of that transformation.

The *greening of an occupation* denotes a structural change in its task composition over time, whereby environmentally beneficial (“green”) tasks become more prevalent relative to white or environmentally harmful (“brown”) tasks. This concept highlights the dynamic nature of occupational content: even jobs not traditionally classified as “green” may increasingly include sustainability-related responsibilities, such as energy efficiency, waste reduction, or sustainable resource use. The GOJI framework measures greening as a shift in the task-level index for a given occupation across different time points, thus quantifying how jobs evolve in response to ecological and technological change.

2.3 International Approaches and Definitional Debates

The delineation of green vs. brown tasks in my framework intersects with several strands of international research and policy on green jobs. It is useful to compare my task-based definitions with other prevalent approaches:

Output-based definitions. A common method, used by statistical agencies and early academic work, defines green jobs by the output of the firm or sector. The U.S. BLS, for example, in its Green Jobs Initiative (2010–2013) counted as green any job in an

establishment that produces goods or services with an environmental benefit (renewable energy, pollution mitigation, recycling, etc.) (Elliott et al., 2024). However, this broad categorization also includes roles that may only indirectly contribute to sustainability, raising concerns about potential overestimation of green employment. Studies such as those by Vona et al. (2018) emphasize that many existing green job classifications rely too heavily on industry-level definitions, which fail to capture task-level environmental relevance. These limitations have prompted a growing body of work focusing on occupational and task-based metrics to understand the greenness of work more accurately. International Labour Organization (ILO) (2016) and OECD (2022) similarly argue for more granular approaches, emphasizing the need to measure not just whether a firm operates in a green industry, but whether individual tasks actually contribute to environmental objectives. In my study, I address this measurement gap by moving beyond output-based classifications to a task-level framework that better reflects the evolving content of work.. This “output approach” essentially designates entire industries or sub-industries as green and counts employment in them. Many national statistical offices in Europe followed a similar approach by measuring employment in the Environmental Goods and Services Sector (EGSS), which includes industries like renewable energy, waste management, and environmental consulting(Eurostat, 2016). While straightforward, the output-based approach has a binary character and can overlook green activities occurring in “non-green” industries. For example, a manufacturing company producing cars might implement cutting-edge energy-saving processes in its factories; output-based metrics would still count all its jobs as non-green because cars are not an environmental good per se.

Process-based and hybrid definitions. To capture environmentally beneficial activities outside the dedicated green sectors, some definitions include a *process perspective*. The BLS extended definition explicitly recognized this, adding a category for jobs involving environmentally friendly production processes and practices within any establishment(Vona et al., 2018)(Dierdorff et al., 2009). These could include roles like sustainability managers, energy auditors, or any workers tasked with improving a firm’s internal

environmental performance (regardless of the firm’s main product). [Sommers \(2013\)](#) describes this as identifying establishments that use processes which “*conserve natural resources or reduce pollution.*” Internationally, the ILO also distinguished between employment in environmental *outputs* and employment in environmental *processes* when defining “employment in environmental activities” ([Dierdorff et al., 2009](#)). Moreover, the ILO introduced an additional criterion that a “green job” should be a decent job – i.e., providing adequate wages, rights, and social protection – arguing that sustainability has both environmental and social dimensions ([Vona et al., 2018](#)). Most statistical implementations (including ours) set aside the decent work criterion as it pertains more to job quality than environmental content.

My task-based approach can be seen as a refinement of the process-based perspective. Rather than classifying whole jobs as process-based green jobs (which could be ambiguous), I zoom in to identify which tasks within a job are related to environmental processes. In effect, I break each occupation into its constituent parts and evaluate those parts. This resonates with what some researchers call a “bottom-up” approach ([Vona et al., 2018](#)), identifying green activities at the micro level within firms or occupations, instead of labeling at the macro level of entire occupations or sectors.

Task-based approaches in research. A growing body of academic work has also employed task data to identify green jobs or the greenness of jobs. Early contributions in this vein include [Dierdorff et al. \(2009\)](#), who were among the first to systematically incorporate the concept of green tasks. Using the O*NET database of occupational descriptors in the United States, they categorized occupations based on how the emerging green economy affects them. Specifically, they introduced categories of occupations such as *Green Increased Demand* (existing occupations that will see rising employment due to green trends), *Green Enhanced Skills* (existing occupations that will significantly change in task content and skill requirements due to greening, though the core work stays the same), and *Green New and Emerging* (entirely new occupations created by the green economy) ([Vona et al., 2018](#)). This taxonomy underscored that greening can reshape the task composition of jobs, even if the job itself does not vanish or newly appear.

[Consoli et al. \(2016\)](#) took a different angle by asking: in what ways do green jobs differ from others? Using U.S. O*NET data and employment records, they found that green jobs tend to require more formal education and high-level cognitive skills. Interestingly, they also found green jobs often demand greater *interpersonal* skills, hypothesizing that many green occupations (like environmental consulting, project management for sustainability initiatives) involve coordinating and communicating across disciplines, hence the need for teamwork and communication skills. Their results, published in *Research Policy*, suggested that green jobs are generally high-skilled and knowledge-intensive roles, dispelling a notion that green jobs might all be “blue-collar” installation work. At the same time, they did identify a spectrum: some green jobs are relatively low-skill (e.g., basic recycling sorters) but many are high-skill (engineers, managers). This interpretation is supported by [Vona et al. \(2018\)](#), who found only modest skill gaps between green and brown jobs, with green occupations requiring slightly more technical expertise. In addition, [Bowen et al. \(2018\)](#) emphasize that green jobs often exhibit greater task complexity, further reinforcing their generally knowledge-intensive nature.

Building on that, [Vona et al. \(2018\)](#) combined O*NET task measures with U.S. labor statistics to examine skill differences between green and brown occupations. They concluded that, contrary to some expectations, the overall skill gap between green and brown jobs is small([Consoli et al., 2016](#)). Green jobs did not uniformly require dramatically different skills than brown jobs; rather, they mostly demanded some additional technical competencies (for example, knowledge of environmental regulations or specific clean technologies) on top of a similar base of general skills([Vona et al., 2018](#)). In other words, a worker in a brown occupation often has a skill set that is largely transferable to a green occupation, needing only targeted upskilling ([Vona et al., 2018](#)). This is an important insight for worker transition policies. Similarly, [Vona et al. \(2019\)](#) find that green employment in the U.S. is concentrated among higher-skilled, higher-wage workers and in certain regions, but again these green workers are not fundamentally a different breed—they are often drawn from adjacent occupations with overlapping skill requirements. Similarly, [Vona et al. \(2019\)](#) find that green employment in the U.S. is

concentrated among higher-skilled, higher-wage workers and in certain regions, but again these green workers are not fundamentally a different breed—they are often drawn from adjacent occupations with overlapping skill requirements(Vona et al., 2019).

The GOJI approach inherits the spirit of these task-based analyses but with a dynamic twist: I apply it to a time series of occupational descriptions (in Germany) that are updated regularly. This allows me not just to classify which occupations are green or brown, but to observe how their task composition shifts year by year. Such within-occupation changes can be significant; indeed, Freeman et al. (2020) (analyzing U.S. data) found that changes in task content *within occupations* accounted for a major share of overall changes in job task requirements economy-wide over recent decades. In the context of environmental change, capturing within-occupation greening is crucial, since much of the transition involves existing jobs evolving rather than entirely new job titles appearing overnight.

Discussion of white tasks. One might ask: if an occupation has mostly white tasks and a small fraction of green or brown tasks, should it be considered a green or brown job at all? This question highlights the nuance introduced by my continuous measure. Traditional approaches might force a cutoff (e.g., an occupation is labeled green if more than 50% of its tasks are green, or if it belongs to a green sector). In contrast, I embrace the continuum. Many occupations will indeed have GOJI values near zero—meaning they have mostly white tasks. That does not render the concept moot; rather, it reflects reality. A large portion of the labor market (e.g., many service and administrative jobs) does not directly interact with environmental objectives. These jobs are “white” in my scheme. The GOJI allows me to identify which of those white-task occupations might be moving in a greener or browner direction. For example, consider a facilities manager: historically their tasks might all be white (managing building operations). If today that job starts to involve overseeing an energy efficiency retrofit (a green task), the GOJI will capture a shift from 0 to a small positive number. This granularity is valuable for pinpointing subtle diffusion of green practices across the economy.

Critique from a complementary perspective. A recent JRC working paper by Villani, González Vázquez, and Fernández-Macías (2025) presents a compelling critique of occupational and task-based frameworks such as the GOJI. The authors argue that relying on occupational titles and predefined task lists can lead to both false positives (i.e., non-green jobs misclassified as green) and false negatives (i.e., environmentally relevant jobs not identified as green). These issues arise due to outdated classifications, inconsistent application across countries, and the erroneous assumption that task content remains stable over time (Villani et al., 2025). Based on empirical data from 24 European countries, they find that the share of green employment—when measured through task-based proxies such as O*NET—has not significantly increased over time and shows only a weak correlation with actual environmental outcomes. This suggests that task-based indicators may produce misleading conclusions if interpreted in isolation.

While the GOJI provides a continuous and occupation-level indicator of environmental task potential, it is not immune to the methodological challenges outlined by Villani et al. To mitigate these limitations, I employ several quality-assurance measures. First, the requirements catalog of the German BERUFENET system is continuously updated. New tasks are evaluated by a dedicated editorial team at the Federal Employment Agency, and environmentally relevant tasks are marked and visually highlighted using a green-leaf symbol. Second, all green and brown tasks undergo a keyword-based validation process at regular two-year intervals to ensure that classification remains accurate and internationally comparable. Third, where available, I triangulate GOJI scores with firm- and sector-level environmental indicators—such as the presence of environmental management systems, environmental attitudes of firms, and the CO₂ intensity of sectors. Finally, I emphasize that GOJI values should be interpreted as potential indicators rather than definitive measures of ecological impact. They must be contextualized in relation to firm-level practices, sectoral dynamics, infrastructure, and the actions and attitudes of individual workers. These enhancements are intended to strengthen the interpretive validity of GOJI while addressing the concerns raised in the broader methodological debate.

In summary, the conceptual foundation of GOJI rests on viewing green labor through a task lens. This approach harmonizes with international calls to incorporate both output and process measures of green employment (Apostel and Barslund, 2024, Wang and Li, 2025), and it operationalizes those ideas at the micro level. It moves beyond the dichotomy of “green job / not green job” and instead acknowledges shades of green (and brown) within each job. As the next section will show, this concept has been implemented in successive versions of my index, progressively refining how tasks are identified and quantified as green or brown.

3 Data Sources and Methods

In this section, I describe the data source and the technical steps involved in constructing the Greenness-of-Jobs Index (GOJI) 4.0. I start with an overview of the BERUFENET occupational database and how it provides task information. I then detail the classification procedure used to identify green and brown tasks—including the official BA Green Skills list (GOJI 3.0 onward), the text-mining-based quality check, and the curated brown dictionary—with illustrative examples. Finally, I present the formulae for computing the index and discuss real occupation examples to demonstrate the calculation.

3.1 Data Source: BERUFENET Occupational Descriptions

My analysis relies on textual descriptions of occupations from **BERUFENET**, which is an online database of occupations maintained by the German Federal Employment Agency. BERUFENET is essentially a comprehensive repository of job profiles, containing information such as job titles, brief descriptions, typical tasks and duties, required qualifications, knowledge and skills, etc. For the purposes of GOJI, the key component of BERUFENET is the *requirements catalog* (“Anforderungskatalog”) of each occupation, which lists the tasks and activities associated with that occupation.

Note on the interpretation of task classification. It is important to emphasize that the GOJI classification reflects the *potential* environmental impact of occupational

tasks. The actual ecological effect of a job depends on multiple contextual factors, including the environmental footprint of the firm or sector, the nature of the goods and services produced, the energy and resource efficiency of production processes, the means of transport and logistics used, the sustainability of regional infrastructure, and finally the personal attitudes and actions of individual workers. A task may be coded as 'green' because it involves environmentally relevant objectives (e.g., energy management), but its impact will vary significantly depending on how it is implemented in practice.

Occupations in BERUFENET are categorised according to the German Classification of Occupations (KldB2010) in a hierarchical five-digit code structure. A five-digit KldB code — e.g. “34332” for *Abfallwirtschaft – Spezialist* (waste management specialist) — identifies an occupational unit (*Berufsgattung*); the KldB2010 contains approximately 1,300 such units. BERUFENET maps these to around 4,600 more granular single occupations (*Einzelberufe*, 8-digit DKZ) that are the actual units of task description.

Each occupation entry distinguishes *core tasks* (essential duties) from *additional tasks* (supplementary duties). For the GOJI construction I combine both categories, retaining a count of the total. BERUFENET is a living database: the Federal Employment Agency revises occupational descriptions to reflect new technologies and changing work practices, though not all occupations are updated every year. I extract annual snapshots from 2012 through 2025, capturing how task content evolves over time. Each snapshot covers approximately 1,200–1,300 occupational units; the 2025 requirements catalog contains around 7,671 distinct task statements (Kuai et al., 2025), while individual occupations average roughly 20 task assignments.

All descriptions are in German, and all pattern matching is applied to German text. Technical terms and compound words specific to German occupational language (e.g. *Fotovoltaikanlagen*, *Hochofentechnik*) are reflected directly in the keyword patterns; English translations in this paper are provided for readability only.

The richness of this data allows a fine-grained measure like GOJI to be feasible. Without such a database, one would have to rely on occupational titles or sector classifications as proxies for greenness, which are far less precise. For example, the BERUFENET de-

scription for *Gärtner/in - Garten- und Landschaftsbau* (Gardener - Landscaping) includes tasks such as “Plan and construct sustainable green spaces” (which might be flagged as green if it includes words like “sustainable”) and “Operate construction machinery for earthworks” (which might be white or brown depending on wording). This level of detail is what I exploit.

A note on scope: GOJI is computed for occupations, not individual jobs or workers. It does not tell me which specific worker is in a green job, but rather how green the typical job in a given occupation is. To relate GOJI to actual employment, one can combine it with employment data by occupation (which I do in some analyses, for instance computing the share of total employment that is in green tasks, etc.). But the construction of GOJI itself is purely on the occupation description side.

3.2 Identifying Green and Brown Tasks: Classification Procedure

The classification of individual tasks as green, brown, or white has evolved across GOJI versions. In GOJI 1.0 and 2.0, both green and brown tasks were identified by applying keyword and regular-expression (regex) patterns to the German text of BERUFENET task descriptions; Appendix B details the algorithm. Starting with GOJI 3.0, green-task identification switched from text-mining to the official BA Green Skills list, while brown-task identification retained a keyword-based approach (yielding 306 brown tasks in GOJI 3.0). GOJI 4.0 completes the transition: brown tasks are also moved from a live text-mining output to a curated allow-list of exact task identifiers, reducing the over-inclusion inherent in keyword matching. Both classifications thus rest on authoritative, reqid-based sources in GOJI 4.0, following the principle of using structured, task-level descriptors advocated by the ILO and Cedefop ([International Labour Organization and European Centre for the Development of Vocational Training \(Cedefop\), 2012](#)).

Green task identification (GOJI 3.0 and 4.0). The primary source is the **BA Green Skills list**, published by the Federal Employment Agency on 24 November 2023.

The list enumerates 378 tasks by their unique requirement identifier (**reqid**) in the BERUFENET requirements catalog. Classification is by exact **reqid** match: a task is green if and only if its identifier appears on the list; no text matching is applied to the task description itself. Representative task examples are given in Appendix A.1.

The keyword and regex patterns described in Appendix A.1 are retained as a *quality-check (QC) framework*: after the reqid-based classification, I verify that pattern-based matching broadly agrees with the list assignments. Systematic disagreements surface tasks that may warrant manual review (e.g. a task containing clear green vocabulary that has not yet been added to the BA list, or vice versa). This QC step does not alter the primary classification.

Brown task identification (GOJI 4.0). Brown tasks are identified through a **curated brown dictionary**: an explicit list of 190 task requirement identifiers selected on the basis of sector-level greenhouse-gas footprint evidence (IPCC, IEA, FAO, and related sources). Classification is by exact **reqid** match. The dictionary was compiled in two stages: first, regex patterns applied to BERUFENET task descriptions produced a candidate set (terms relating to fossil fuels, energy-intensive industrial processes, livestock, aviation, etc.); second, this set was manually reviewed to remove false-positive matches and, where supported by sectoral GHG evidence, to add further tasks. The final list is a static curated allow-list, not a live text-mining output. This design eliminates the over-inclusion characteristic of GOJI 3.0’s text-mining approach, in which compound-word and partial-string matches produced ambiguous classifications (e.g. *Goldschmiedearbeiten* matching via *Schmied*, or *Käseherstellung* matching via *Milch*). The full reqid-level change table between GOJI 3.0 and GOJI 4.0 is in Appendix B. Representative brown-task examples are in Appendix A.2.

White task identification. Any task in the requirements catalog that matches neither the BA Green Skills list nor the curated brown dictionary is classified as **white**.¹ White

¹Earlier GOJI versions — in particular GOJI 2.0 (v1220) — used the term *neutral* for this third task category. GOJI 3.0 and 4.0 adopt *white* throughout, in line with the three-colour framework (green/brown/white).

tasks constitute the large majority of all task assignments — roughly 94% of distinct tasks in the 2025 snapshot (Appendix A.3).

No green–brown overlap (GOJI 4.0). Because both classifications are reqid-based allow-lists, a task cannot appear in both simultaneously. The priority rule required in keyword-based approaches (“if green and brown keywords both match, assign green”) is therefore not needed in GOJI 4.0.

Temporal constancy of the task classification. Within a given GOJI version, the green and brown task lists are held *constant across all years in the panel*. The same classification applied to the 2025 BERUFENET snapshot is applied retroactively, without modification, to every annual snapshot from 2012 onward. Consequently, year-on-year changes in GOJI within a version reflect solely changes in the occupation–skills matrices — that is, which tasks appear in an occupation’s BERUFENET profile and with what status (core versus additional) — not any reclassification of individual tasks.

Sources of differences across GOJI versions. Differences in published GOJI values between versions arise from three sources. *(i) Classification changes:* the set of tasks designated green or brown may differ between versions; because the classification is applied retroactively to all years, any revision to the task lists shifts GOJI values for the entire historical panel simultaneously. *(ii) Task-composition changes in the occupation–skills matrices:* BERUFENET occupation profiles for past years do not change after their initial annual release; therefore, changes in task composition at the 8-digit single-occupation level typically emerge only when new annual snapshots extend the time series, not from retrospective edits. *(iii) Occupational-composition changes at the 5-digit level:* the mapping from 8-digit single occupations to 5-digit occupational units (KldB 2010 *Berufsgattungen*) may change between versions if single occupations are created, revised, or retired in the BERUFENET population; this can shift the employment-weighted GOJI at the 5-digit level even when no task classification and no individual occupation profile has changed. Appendix E documents the quantitative effect of each source across GOJI

versions 1.0–4.0.

Computing occupation-level GOJI. For each occupation o in year t , after classifying all tasks:

$G_{o,t}$ = number of green tasks, $B_{o,t}$ = number of brown tasks, $T_{o,t}$ = total tasks (core + additional)

The positive and negative greenness-of-jobs indices are:

$$GOJI_{o,t}^+ = \frac{G_{o,t}}{T_{o,t}}, \quad (1)$$

$$GOJI_{o,t}^- = \frac{B_{o,t}}{T_{o,t}}. \quad (2)$$

The net greenness index is:

$$GOJI_{o,t} = GOJI_{o,t}^+ - GOJI_{o,t}^- = \frac{G_{o,t} - B_{o,t}}{T_{o,t}}.$$

This net index is the share of tasks that are green minus the share that are brown.

To ensure clarity, let’s work through three examples drawn from real GOJI 4.0 data.

Dark-green example: Roofer (*Dachdeckerei* – *Fachkraft*, KldB 32142). The roofer’s BERUFENET profile includes tasks such as installing and maintaining photovoltaic and solar-thermal roof systems, applying thermal-insulation layers (green roof, insulating facades), and complying with energy-efficiency standards. These trigger multiple green-list entries. The profile also contains a small number of tasks related to conventional roofing materials without environmental implications. No brown-list keywords are matched. In GOJI 4.0 (2024), $GOJI_{32142,2024}^+ \approx 0.244$, $GOJI_{32142,2024}^- \approx 0.005$, net $GOJI \approx +0.238$. The roofer is classified as dark green. Over time, its net GOJI rose from +0.113 in 2012 to +0.238 in 2025 as solar and insulation tasks became more prominent in the profile.

Dark-brown example: Construction-materials production (*Baustoffherstellung – Fachkraft*, KldB 21222). This occupation involves producing concrete products, cement-based materials, and related construction inputs. Tasks such as operating kilns, handling concrete mixes, and controlling sintering processes trigger numerous brown-list entries (cement, combustion, energy-intensive materials). Green-task matches are negligible (a small environmental monitoring component). In GOJI 4.0: $GOJI_{21222}^+ \approx 0.006$, $GOJI_{21222}^- \approx 0.247$, net $GOJI \approx -0.241$. Dark brown.

White-task example: Health and nursing (*Gesundheits- und Krankenpflege (o.S.) – Fachkraft*, KldB 81302). This occupation encompasses general nursing tasks: patient care, monitoring vital signs, administering medication, documentation. None of these match either the green list or the brown dictionary. In GOJI 4.0: $GOJI^+ = 0$, $GOJI^- = 0$, net $GOJI = 0.000$ in every year 2012–2025 (employment: 570,000–669,000 workers). This provides a clean real-world illustration of a large, stable occupation with exclusively white tasks.

These real examples highlight how GOJI works at the occupation level: a clear sectoral logic (roofing $\rightarrow$ solar; cement $\rightarrow$ combustion; nursing $\rightarrow$ white/zero) is confirmed by the task-list evidence. For GOJI 3.0 and 4.0, the primary green signal comes from the official BA Green Skills list, so misclassification risk on the green side is much lower than in text-mining-only approaches; the brown classification and the QC step for green tasks still rely on the careful crafting of regex patterns and iterative manual checks.

3.3 Calculation of the Index and Aggregation

Once $GOJI_{o,t}$ is computed for all occupations o , the index can be analysed and aggregated in several ways. The distribution of GOJI across occupations in a given year clusters around zero with long tails, as shown in Appendix C. Weighting each occupation’s index by its employment share yields an economy-wide greenness-of-employment measure that tracks the share of tasks in the entire economy that are green (versus brown); [Janser \(2018, 2019\)](#) document an upward trend in this measure for Germany.

Aggregation to higher geographical or sectoral levels follows the same employment-weighting principle. Let $L_{o,s,t}$ denote employment of occupation o in sector s in year t (from linked administrative employment records). Total employment in sector s is $L_{s,t} = \sum_{o \in s} L_{o,s,t}$, and the employment-weighted sector-level GOJI is

$$GOJI_{s,t} = \sum_{o \in s} \frac{L_{o,s,t}}{L_{s,t}} \cdot GOJI_{o,t}.$$

I apply this scheme separately at each level of the WZ08 classification (three-, two-, and one-digit) and also across KldB2010 occupational groups, requirements levels, and administrative regions. Sections 4–6 present the results of these aggregations. The same formula applies to the positive and negative components $GOJI_{o,t}^+$ and $GOJI_{o,t}^-$ when green and brown task shares at higher levels are required.

The index can also be merged with individual-level administrative data to study correlations between occupational greenness and wages, education levels, or demographic composition. [Bowen et al. \(2018\)](#) and [Kuai et al. \(2025\)](#) document a modest green wage premium; [Consoli et al. \(2016\)](#) find that green occupations tend to require more formal education; [Vona et al. \(2019\)](#) note that green jobs are disproportionately held by men.

GOJI measures *potential environmental task content* as recorded in official occupational profiles, not actual environmental performance. This distinction is important: the inclusion of a green task in a BERUFENET profile indicates that the activity is part of the expected role, but the ecological impact depends on how it is implemented in practice and on the broader sectoral and technological context.

Employment statistics of the Federal Employment Agency (BA) are available at the 5-digit KldB2010 level only from 2013 onwards.² All employment-weighted aggregations therefore begin in 2013. The 2013 and 2025 snapshots serve as the main comparison points in Sections 4–6 below.

²The 5-digit KldB2010 and 8-digit DKZ occupation-level GOJI indices, which do not rely on employment statistics, are available from 2012.

4 Aggregation to Higher Occupational Levels

Table 1 compares employment-weighted net GOJI across the 14 KldB2010 occupational segments (*Berufssegmente*) in 2013 and 2025, disaggregated by requirements level (1 = Helfer through 4 = Experte). Figure 1 shows the full 2013–2025 time series of brown, green, and net GOJI for each segment.

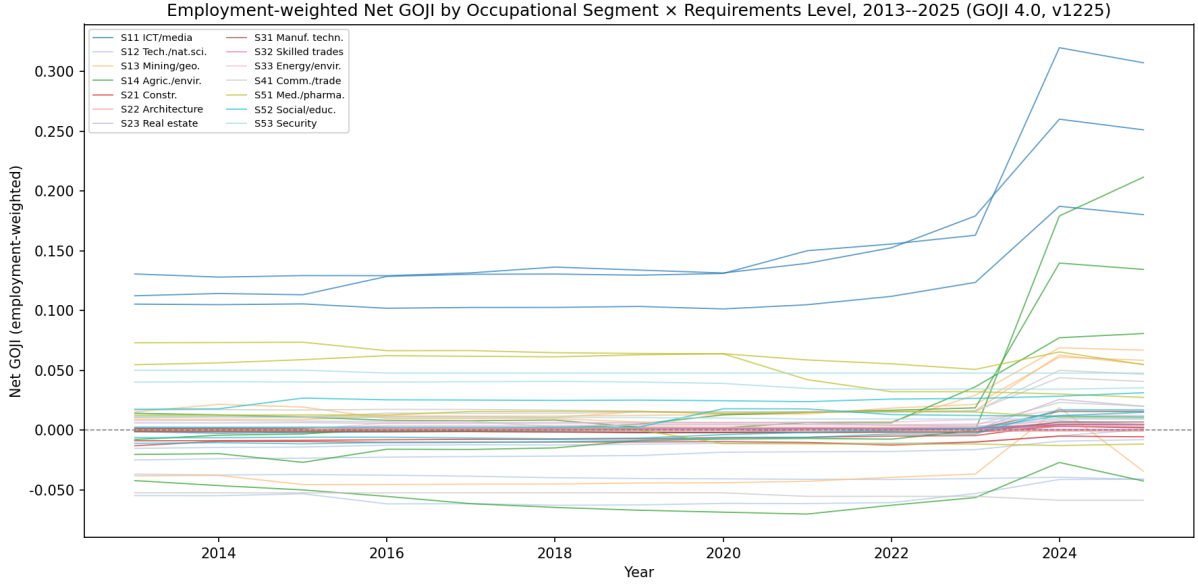


Figure 1: Employment-weighted net GOJI by occupational segment and requirements level, 2013–2025 (GOJI 4.0, v1225). Each line represents one segment $\times$ requirements-level cell. **Data sources:** GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi, KldB2010 5-digit).

5 Aggregation to Industry Level

The sector-level GOJI is constructed by employment-weighting occupation-level values within each industry cell (see Section 3.3 for the formula). I apply this scheme to the full hierarchy of the WZ08 classification: three-digit industries, two-digit divisions, and one-digit sections (labelled A–U; manufacturing = Section C). All employment-based aggregations begin in 2013.

Illustrative example: WZ08 261. To illustrate how individual occupation movements translate into sector-level GOJI, Table 2 reports values for *Mechatroniker/in* –

Fachkraft (KldB 26112), a representative occupation in WZ08 261 (*Herstellung von elektronischen Bauelementen*, electronic components manufacturing).

Table 2: GOJI Values for *Mechatroniker/in – Fachkraft* (KldB 26112), 2013 and 2024

Year	GOJI ⁺ (green)	GOJI [−] (brown)	Net GOJI
2013	0.000	0.000	0.000
2024	0.182	0.000	+0.182
Δ 2013–2024	+0.182	0.000	+0.182

The example illustrates the step-change visible in many technical occupations after the BA’s BERUFENET green-leaf rollout: this occupation had no classified tasks of either colour from 2012 through 2022 (net GOJI exactly 0 throughout), then shifted sharply to dark green in 2024 once its profile was updated with green-leaf-marked tasks such as maintaining electric-drive systems and regenerative-energy equipment.

Table 3 illustrates how this occupation-level shift aggregates upward. Let $L_{o,261,t}$ denote the number of employees in occupation o working in WZ08 261 in year t , and let $L_{261,t} = \sum_{o \in 261} L_{o,261,t}$ denote total employment in WZ08 261. The employment share of occupation o within WZ08 261 is

$$w_{o,261,t} = \frac{L_{o,261,t}}{L_{261,t}},$$

and the three-digit sector-level index is defined as

$$GOJI_{261,t} = \sum_{o \in 261} w_{o,261,t} \cdot GOJI_{o,t}.$$

The same weighting logic applies for aggregations to the two-digit and one-digit WZ08 levels. At the one-digit level, the WZ08 classification uses letter sections (A–U); manufacturing is Section C. Both indices are constructed as employment-weighted means of the relevant lower-level group indices.

Table 3: Net GOJI (employment-weighted) for WZ08 261, WZ08 26 and WZ08 Section C, 2013 and 2024 (GOJI 4.0, v1225)

Year	WZ08 261	WZ08 26	WZ08 Section C
2013	+0.0059	+0.0038	−0.0108
2024	+0.0281	+0.0272	+0.0108
Δ 2013–2024	+0.0222	+0.0234	+0.0216

WZ08 261 (electronic components) was already mildly green-positive in 2013. Aggregating to WZ08 26 (computing, electronics, and optics) yields a slightly lower net GOJI because the two-digit sector contains more mixed occupations. At the one-digit level (Section C: manufacturing), net GOJI was slightly negative in 2013 but turned positive by 2024, driven by the green-leaf rollout in technical occupations.

All WZ08 one-digit sections, 2013–2025. Table 4 presents employment-weighted net, green, and brown GOJI for all WZ08 one-digit sections in 2013 and 2025, with the change $\Delta_{2013-2025}$. Figure 2 shows the corresponding time series from 2013 to 2025.

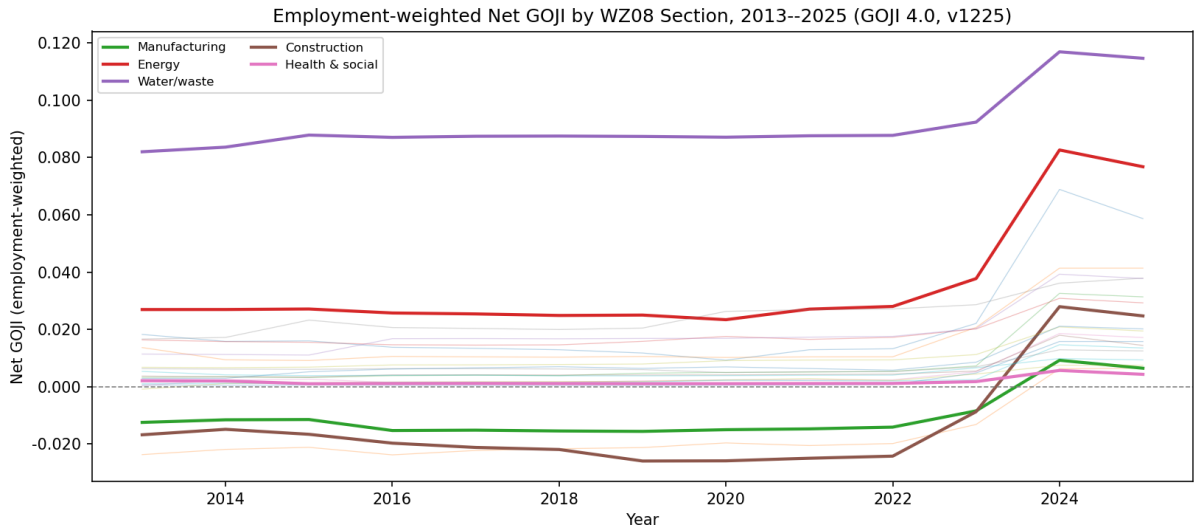


Figure 2: Employment-weighted net GOJI by WZ08 one-digit section, 2013–2025 (GOJI 4.0, v1225). Selected sections labelled; remaining sections shown in grey. **Data sources:** GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi, KldB2010 $\times$ WZ08).

6 Aggregation to Regional Level

The employment-weighted GOJI can be aggregated to administrative regions using the geographic distribution of employment as weights, following the same formula as in Section 3.3. Figures 3–5 show brown, green, and net GOJI at the level of the 16 German federal states (*Bundesländer*) for 2013 and 2025. Figures 6–7 drill down to the district (*Kreis*) level (NUTS-3, $n \approx 401$) and show the net index cross-sectionally and its change over the full period.

Federal states (Bundesländer)

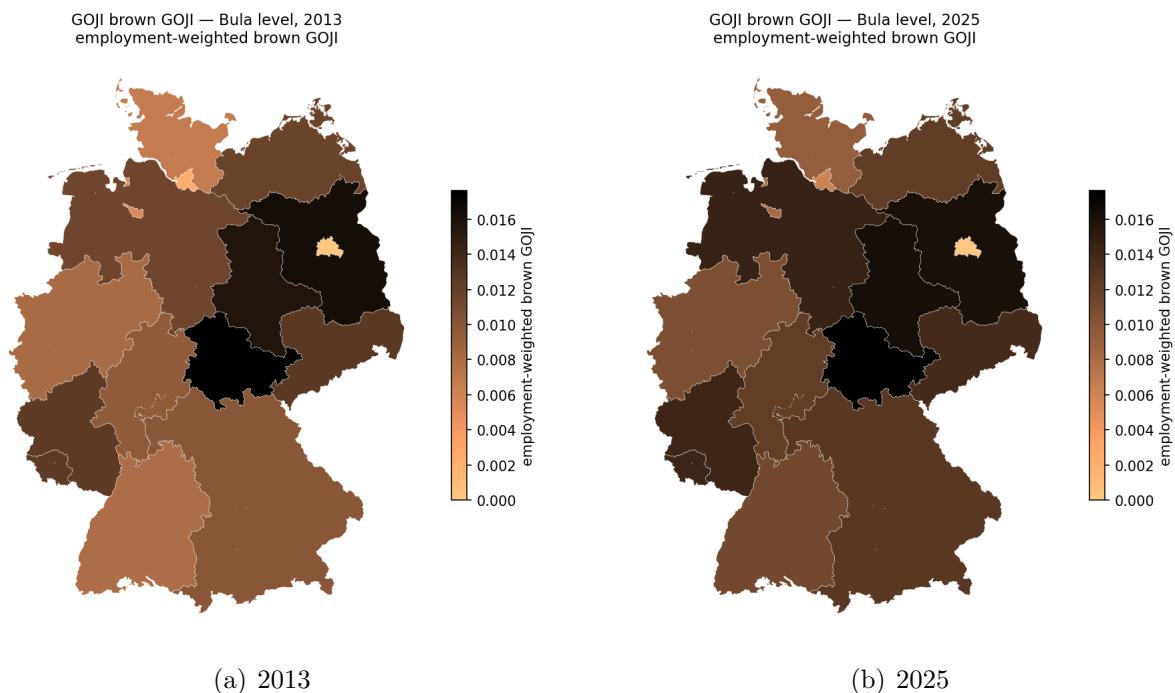


Figure 3: Employment-weighted brown task share (GOJI^-) by federal state, 2013 and 2025. Darker shading indicates higher brown task intensity. Source: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi).

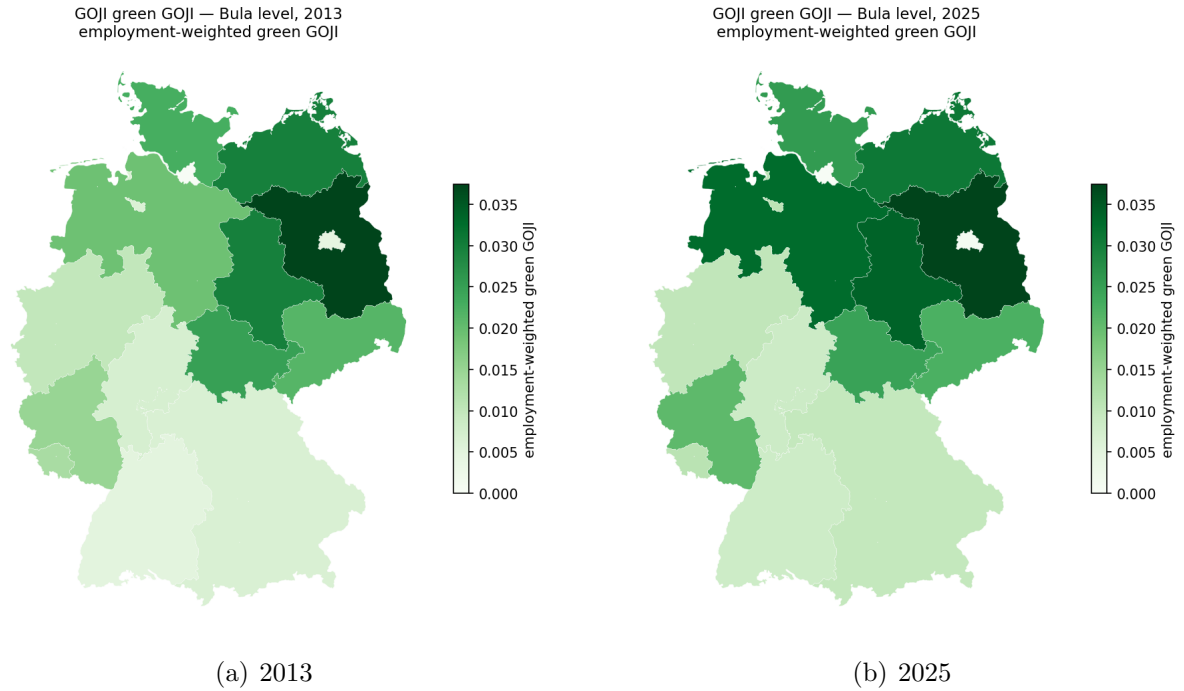


Figure 4: Employment-weighted green task share (GOJI⁺) by federal state, 2013 and 2025. Darker shading indicates higher green task intensity. Source: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi).

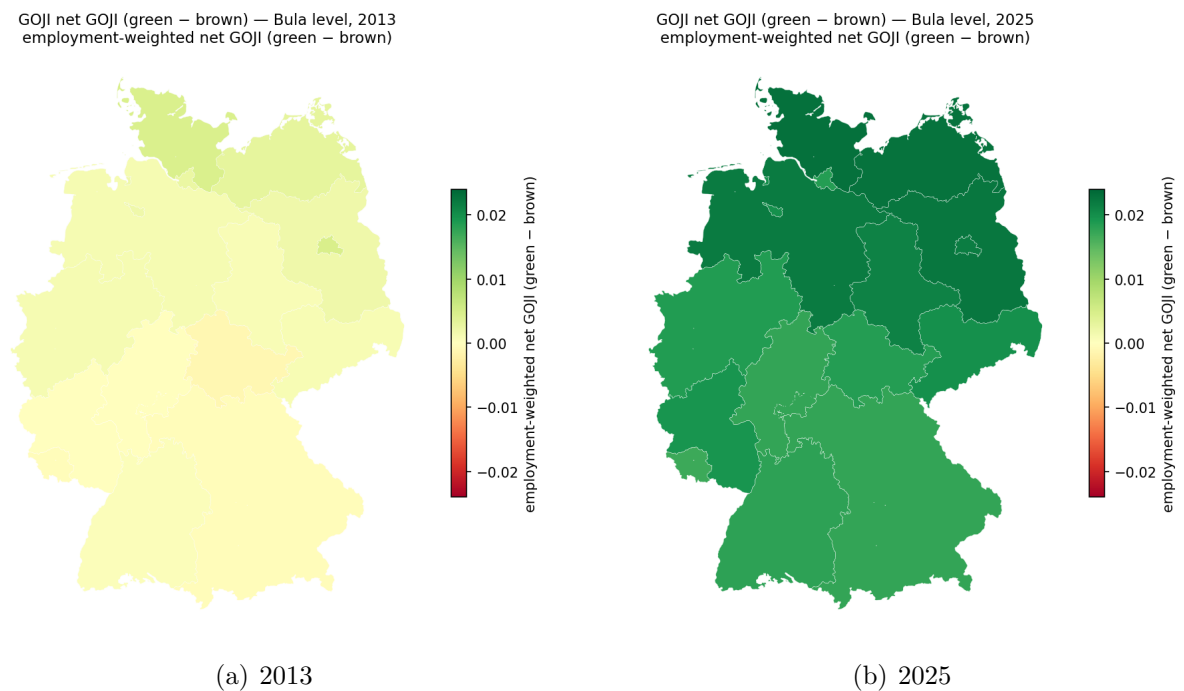


Figure 5: Employment-weighted net GOJI by federal state, 2013 and 2025. Green (red) shading indicates net-positive (net-negative) occupational task content. Source: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi).

GOJI net GOJI (green – brown) — Kreis level, 2013
employment-weighted net GOJI (green – brown)

GOJI net GOJI (green – brown) — Kreis level, 2025
employment-weighted net GOJI (green – brown)

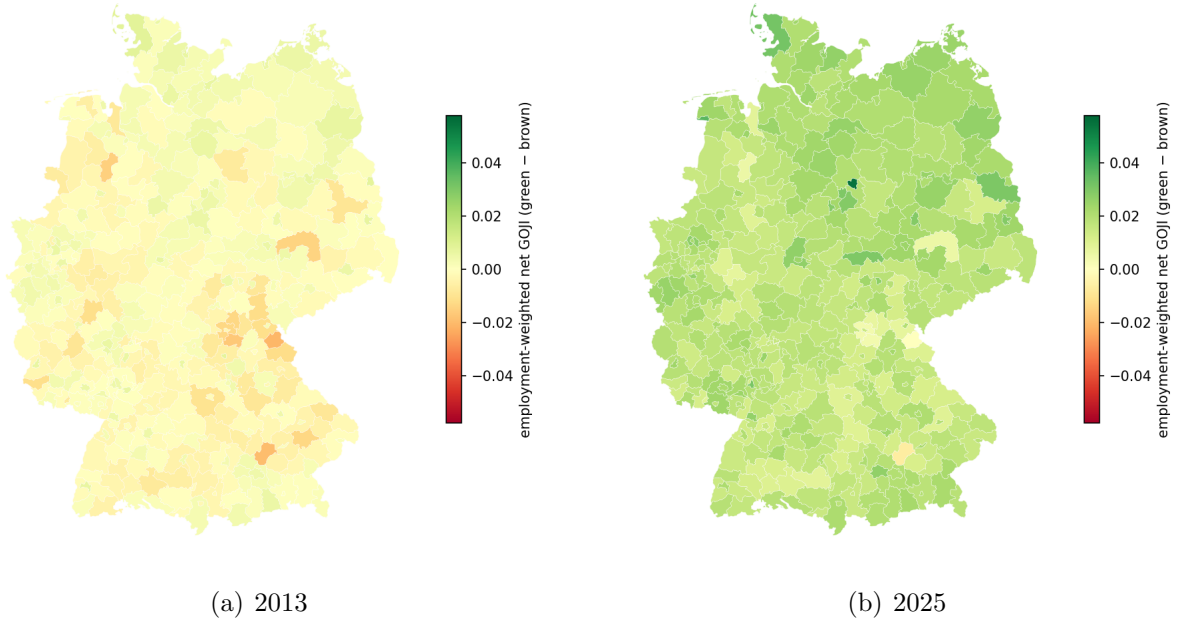


Figure 6: Employment-weighted net GOJI by district (*Kreis*, NUTS-3), 2013 and 2025. Green (red) shading indicates net-positive (net-negative) task content. Source: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi).

Districts (Kreise, NUTS-3)

7 Conclusion

The Greenness-of-Jobs Index (GOJI) offers a fine-grained, continuously updated measure of the environmental task content of occupations in Germany. By classifying tasks rather than whole jobs or industries, GOJI captures the full spectrum from dark-green to dark-brown and tracks how individual occupations evolve as technologies and regulations change. The current version, GOJI 4.0 (v1225), covers the years 2012–2025 and is the first to make the BA’s green-leaf rollout empirically visible as a broad-based rise in green task shares across technical and engineering occupations.

Greenness has spread across the labour market, but incrementally. Over the past decade many occupations incorporated environmentally relevant tasks and raised their GOJI scores; the employment-weighted net index trends upward throughout the period and accelerates sharply after the 2023–2024 BERUFENET update. Nevertheless, the

GOJI net GOJI (green – brown) — change 2013→2025, Kreis level
 Δ employment-weighted net GOJI (green – brown)

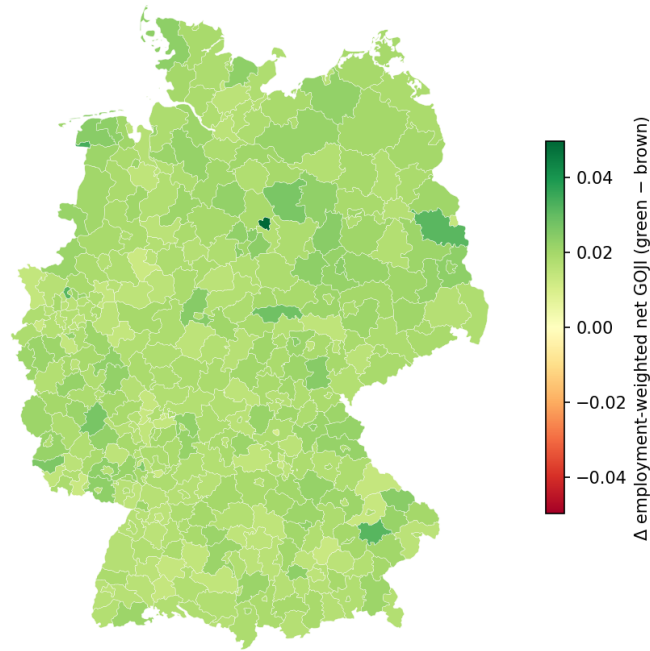


Figure 7: Change in employment-weighted net GOJI by district (*Kreis*, NUTS-3), 2013–2025. Green shading = increase in net greenness; red shading = decrease. Source: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi).

large majority of German employment remains in occupations with predominantly white tasks, implying both a need for continued effort and substantial untapped potential for greening.

Brown tasks remain concentrated in a minority of occupations — fossil-fuel extraction, conventional energy generation, heavy industry, and livestock production — but those occupations collectively account for a non-trivial employment share. GOJI identifies precisely which occupational units and workers are most exposed to transition risk, providing a quantitative foundation for targeting upskilling programmes and transition assistance.

At the aggregate level, greenness is positively associated with requirements level and skill intensity, consistent with international evidence (Consoli et al., 2016, Vona et al., 2018). This implies that the green transition may amplify skill-biased employment trends unless accompanied by targeted training investments. Regional variation, as documented in Section 6, further suggests that the transition’s benefits and costs will not be distributed

uniformly across German federal states.

The development of GOJI 4.0 also provides some methodological lessons. I found that combining expert-driven keyword selection with iterative data-driven refinement is an effective way to classify tasks. While my approach was applied to German data, the concept is transferable. Other countries with rich occupational information (such as Canada’s National Occupational Classification descriptions, or O*NET in the U.S., or ESCO in the EU context) could develop their own GOJI-like indices. Indeed, an interesting avenue for future research is to compare the greenness of jobs internationally: are German occupations greener or browner than their counterparts in other countries? Differences might reflect varying industry composition or policy strictness.

However, one must be cautious. GOJI measures the *intended* or *required* environmental content of jobs as described in official sources. It does not measure actual environmental performance (e.g., whether a company truly reduces emissions as a result of those tasks being present) nor the effort put by individual workers into those tasks. There may be instances of “window dressing” where job descriptions mention sustainability but in practice the role spends little time on it. Conversely, some workers may innovate greener practices that are not yet reflected in formal descriptions. Over time, job profiles will likely evolve to catch up with reality (or vice versa), but a lag can exist.

There is also the matter of tasks that might be environmentally relevant but are not captured by the keyword method due to subtle phrasing. While I have attempted to be comprehensive, language is nuanced and evolving. I anticipate updating the dictionaries periodically as new jargon and technologies emerge (for example, terms around hydrogen economy, carbon capture, etc., might become more common and need inclusion). The structure of GOJI allows for such updates.

Another limitation is that GOJI is inherently a measure at the occupation level, and thus cannot differentiate within an occupation. In reality, not every worker in a given occupation will perform the exact same tasks—there’s variation. For instance, an electrical engineer working for a coal power plant will have a different daily focus than an electrical engineer working for a wind turbine manufacturer, yet they might share

the same occupation code. In such cases, GOJI provides an average. In the future, more granular data like job postings or employee surveys could be used to refine within-occupation variation, but that becomes complex. My stance is that occupational level is a useful and tractable level of analysis for policy and forecasting.

The findings from GOJI 4.0 highlight the importance of continuous workforce upskilling. Many brown tasks can be transformed into green tasks through training. For example, a truck mechanic today (brown-leaning) could be tomorrow’s electric truck mechanic (green-leaning) if provided the right training in electric drivetrains. I see evidence of such transitions in the occupational descriptions themselves: new tasks get appended as technology changes. Facilitating workers to acquire those new tasks is critical. In this sense, GOJI could be used to monitor whether training curricula and vocational programs are updating their content to include these new green tasks. If an occupation’s GOJI is rising but the training programs for that occupation are not yet teaching the corresponding green skills, a gap exists.

I also emphasize that a “just transition” requires not only creating green tasks/jobs but ensuring that the workers from brown jobs can move into them. Literature suggests skill overlap is often sufficient, but practical barriers like geographic mobility, age, or credentialing can still hinder transitions. Policies ranging from job search assistance, relocation support, apprenticeships in green tech for mid-career workers, to broader social safety nets will remain important.

In conclusion, the Greenness-of-Jobs Index provides a powerful tool for analyzing the labor market aspects of the environmental transition. It quantifies an otherwise abstract concept—how green is a job?—in a way that is comparable across occupations, regions, and time. As nations worldwide pursue decarbonization and sustainable growth, tools like GOJI can help track progress and inform policy adjustments. The evidence from GOJI 4.0 is cautiously optimistic: economies can green themselves without massive job losses, but the transformation is slow and requires active support. The workforce of the future will likely look different, with new skills and tasks centered around sustainability. By measuring and understanding these shifts, I can better prepare for them, ensuring

that the green economy works for everyone.

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A Appendix A: Classification of Tasks and Examples

A.1 Green Task Examples: Selected Tasks from the BA Green Skills List

Since GOJI 3.0, green tasks are identified using the official **BA Green Skills list** published by the Federal Employment Agency (Bundesagentur für Arbeit, BA) on 24 November 2023. GOJI 4.0 uses the version dated 19 March 2025, which contains **378 distinct green tasks**. The following examples illustrate the range of occupational tasks covered, organised by thematic domain (German task names as they appear in the BERUFENET requirements catalog).

- **Forestry and nature conservation:** *Bergwälder pflegen und bewirtschaften* (managing mountain forests), *Forstschutz* (forest protection), *Waldpflegearbeiten* (forest maintenance work), *Rekultivierung* (land recultivation).
- **Horticulture and organic agriculture:** *Biologischer Pflanzenschutz* (biological pest control), *Ökologischer Landbau* (organic farming), *Kompostierung* (composting), *Naturgärten* (natural garden design).
- **Water management and nature-based infrastructure:** *Gewässerschutz* (water body protection), *Wasseraufbereitung* (water treatment), *Kanalbau / Kanalsanierung* (sewer construction and rehabilitation), *Dachbegrünung* (green roofing).
- **Sustainable construction:** *Energetische Gebäudedämmung* (thermal insulation of buildings), *Dämmungen herstellen* (producing insulation materials), *Ökologisches Bauen* (ecological construction), *Baubiologie* (building biology).
- **Sustainable transport and energy:** *Öffentlicher Personennahverkehr (ÖPNV)* (public local transport), *Fahrradtechnik* (bicycle technology), *Elektrische Antriebstechnik* (electric drive technology), *Fotovoltaikanlagen* (photovoltaic systems).
- **Environmental management and law:** *Umweltaudit* (environmental audit), *Umweltberatung* (environmental consultancy), *Umweltrecht* (environmental law),

Abfallrecht (waste management law).

The full list of 378 green tasks spans a broad range of occupational domains, from traditional green sectors (forestry, horticulture, water management) to technical and policy-oriented roles. Figure 8 shows a word cloud of the most frequently assigned green tasks, where word size is proportional to the number of distinct single occupations (in German: “Einzelberufe”, 8-digit level of DKZ) in which the task appears in the 2025 BERUFENET snapshot.



Figure 8: Word cloud of green tasks by occurrence frequency (2025 BERUFENET snapshot). Word size is proportional to the number of distinct single occupations assigned the task. Source: own calculations from BA Green Skills list and BERUFENET.

A.2 Brown Task Examples: Selected Tasks from the Curated Brown Dictionary

Brown tasks are identified using a **curated brown dictionary** developed for GOJI. In GOJI 4.0, this dictionary contains **190 task entries** matched by exact requirement identifier. The following examples illustrate the covered themes:

- **Fossil fuel extraction and processing:** *Steinkohlebergbau* (hard coal mining), *Braunkohlebergbau* (lignite mining), *Kohleveredlung* (coal upgrading), *Erdöl-*, *Erdgas-*

gewinnung, -förderung (oil and gas extraction), *Koksherstellung, Kokerei* (coke production).

- **Energy-intensive industrial processes:** *Eisen- und Stahlhüttentechnik* (iron and steel smelting), *Hochofentechnik* (blast furnace operation), *Aluminiumhüttentechnik* (aluminium smelting), *Zement herstellen* (cement production), *Betonfertigteile herstellen* (prefabricated concrete production).
- **Plastics and petrochemical production:** *Kunststoffherstellung* (plastics manufacturing), *Kunststoffverarbeitung* (plastics processing), *Petrochemie* (petrochemicals).
- **Livestock and animal-product processing:** *Fleischerzeugung* (meat production), *Schweinezucht* (pig farming), *Rinderzucht* (cattle farming), *Maschinenmelken* (machine milking).
- **Aviation and combustion-engine technology:** *Luftfahrttechnik* (aeronautical technology), *Triebwerkstechnik* (engine and turbine technology), *Dieselmotoren* (diesel engines), *Verbrennungsmotoren* (combustion engines).
- **Other carbon-intensive industrial processes:** *Gießereitechnik* (foundry technology), *Gesenkschmieden* (die forging), *Torfgewinnung* (peat extraction), *Gussasphalt herstellen* (cast asphalt production).

Figure 9 shows a word cloud of the most frequently assigned brown tasks.

A.3 White Task Examples: Selected Neutral Tasks from the Requirements Catalog

White tasks are those present in the BERUFENET requirements catalog that appear in neither the BA Green Skills list nor the curated brown dictionary. They constitute the large majority of all task assignments. The following real examples are drawn from the 2025 requirements catalog across different professional domains:

Brown tasks (2025) -- frequency = number of occupations

Figure 9: Word cloud of brown tasks by occurrence frequency (2025 BERUFENET snapshot). Word size is proportional to the number of distinct single occupations assigned the task. Source: own calculations from curated brown dictionary and BERUFENET.

White tasks represent the core occupational duties that, while essential to the job, have no direct environmental relevance in either direction. They account for the bulk of all task assignments across every occupation in BERUFENET. Figure 10 shows the 300 most widespread white tasks.



Figure 10: Word cloud of the 300 most widespread white (neutral) tasks (2025 BERUFENET snapshot). Word size proportional to number of distinct single occupations assigned the task. All 8,000+ white tasks are neutral; the top 300 are shown for legibility. Generated by `prog/99_wordcloud_top300.py`. Source: own calculations from BERUFENET requirements catalog.

A.4 Occupation Vignette

To illustrate how tasks combine in a real occupation, Table 5 presents actual GOJI 4.0 task assignments for a selected occupation from the 2025 BERUFENET snapshot, drawn from the GOJI 4.0 (v1225) task panel.

B Appendix B: Methodological Evolution and Technical Details

This appendix documents how the GOJI index was developed across four successive versions and provides technical details on the text-processing algorithm used for quality control. A detailed quantitative side-by-side comparison — including employment-weighted GOJI time series and example occupation values — is in Appendix E.

B.1 B.1 Version History

Developing a robust measure of the greenness of jobs has been an iterative process. Table 6 summarises the four published versions.

GOJI 1.0 (2018). The concept of GOJI was first operationalised by Janser (2018, 2019), who applied text-mining to BERUFENET to identify green tasks. For each occupation o with T tasks in total and G green tasks, the index was $GOJI = G/T$. All non-green tasks were implicitly white; there was no brown measure. The 2018 analysis found a modest upward trend in the employment-weighted greenness index from 2006 to 2016. A limitation was that an occupation with purely environmentally harmful activities and an administrative support occupation would both score zero, despite being fundamentally different in environmental relevance.

GOJI 2.0 (v1220). GOJI 2.0 introduced a brown component. A curated dictionary of keywords relating to fossil fuels and energy-intensive industrial processes was applied to BERUFENET task texts, yielding $GOJI_{o,t}^-$ (brown share). The net index $GOJI_{o,t} = GOJI_{o,t}^+ - GOJI_{o,t}^-$ allowed occupations to be placed on a continuous spectrum from dark-brown (large negative net) to dark-green (large positive net). GOJI 2.0 also introduced the group-level categories 3e and 5e. The employment-weighted category shares in GOJI 4.0 (v1225) data for 2013 are: dark green ($>+0.1$) 3.25 %, light green 12.45 %, white (exactly 0) 67.4 %, light brown 13.43 %, dark brown (<-0.1) 3.48 %. Earlier versions with broader keyword matching reported somewhat higher tail shares.

GOJI 3.0 (v1222) and GOJI 4.0 (v1225). GOJI 3.0 marked the most important methodological turning point: green-task classification shifted from text mining to the official **BA Green Skills list** (published 24 November 2023; applied retroactively to all years). Text mining was retained only as a quality check. Occupation-level imputation was introduced for occupations without BERUFENET profiles. GOJI 4.0 updates the green list (378 tasks) and brown dictionary (190 tasks), migrates the pipeline to Python, and extends coverage to 2025. The 2023–2024 data capture the empirical effect of the

BA’s BERUFENET green-leaf rollout for the first time.

To illustrate the version evolution concretely, consider *Mechatroniker/in – Fachkraft* (KldB 26112). In GOJI 1.0 it scored 0 (no green keywords matched). In GOJI 2.0 and 3.0, net GOJI remained 0.000 in every year 2012–2022 (all-white classification). By GOJI 4.0 (2024 snapshot), the occupation’s green share jumped to +0.182 (dark green), reflecting the green-leaf-marked tasks added to its profile: maintaining electric-drive systems and regenerative-energy equipment.

B.2 B.2 Preprocessing and Pattern Matching (QC Algorithm)

The following steps describe the text-processing algorithm used for the quality-check layer in GOJI 3.0 and 4.0 (green) and for the primary brown classification:

1. **Data ingestion.** Occupational descriptions are structured as (occupation_code, year, task_text) records, one row per task assignment.
2. **Text normalisation.** Each task text is lowercased; German umlauts are kept in Unicode form and handled directly by the regex engine. Punctuation within compound words (hyphens) is preserved.
3. **Regex patterns.** Two lists of regex patterns are compiled (green and brown). Examples: `\benergiespar` matches *energiesparend* (energy-saving); `\bkohle` matches coal-related terms; `Emission(?!.*reduz)` matches “Emission” only when *reduz(ieren)* (reduce) does *not* follow — avoiding false-positive brown classification of emission-reduction tasks.
4. **Matching procedure.** For each task text, green patterns are applied first. A green match marks the task green and no brown test is performed (preventing double classification). If no green pattern matches, brown patterns are applied. If neither matches, the task is white.
5. **Edge cases.** A match anywhere in the task text is sufficient to classify the whole task. Sentence boundaries within long task strings are not treated as separate

classification units.

B.3 B.3 Quality Checks and Iteration

Manual review of random samples across green, brown, and white tasks confirmed that classifications were consistent with domain knowledge. Formal checks included: (i) verifying $G_{o,t} + B_{o,t} \leq T_{o,t}$ for all observation-year pairs; (ii) examining occupations with both $G_{o,t} > 0$ and $B_{o,t} > 0$ (typically transitional technical roles, e.g. automotive engineers straddling conventional and electric drivetrains); and (iii) checking for systematic trend reversals. Net GOJI is generally stable or rising over time for most occupations, with a sharp broad-based rise in the 2023–2024 snapshots and no systematic greening reversal outside those rollout years. Borderline classification cases — such as tasks mentioning *Abgas* (exhaust gas) in an emission-*reduction* context — were resolved by adding context-sensitive regex patterns (e.g. matching *Emissionsminderung* as green rather than allowing the bare *Emission* pattern to fire).

B.4 B.4 Methodological Comparison across Versions

Table 7 summarises key methodological differences. A detailed quantitative comparison is in Appendix E.

C Appendix C: Additional Descriptives

C.1 Continuous Distribution of Net GOJI Across Single Occupations

Figure 11 shows the kernel-density estimate of the net GOJI ($GOJI^{\text{net}}$) across all 8-digit single occupations (DKZ) for selected years 2012, 2018, and 2025. The distribution is sharply peaked at zero in every year, reflecting the large majority of occupations with exclusively white tasks (no green or brown task assignments). Both tails are thin: comparatively few occupations accumulate substantial green shares (right tail) or substantial brown shares (left tail). A modest rightward shift is visible from 2012 to 2025, consistent with the secular greening trend documented in the main text.

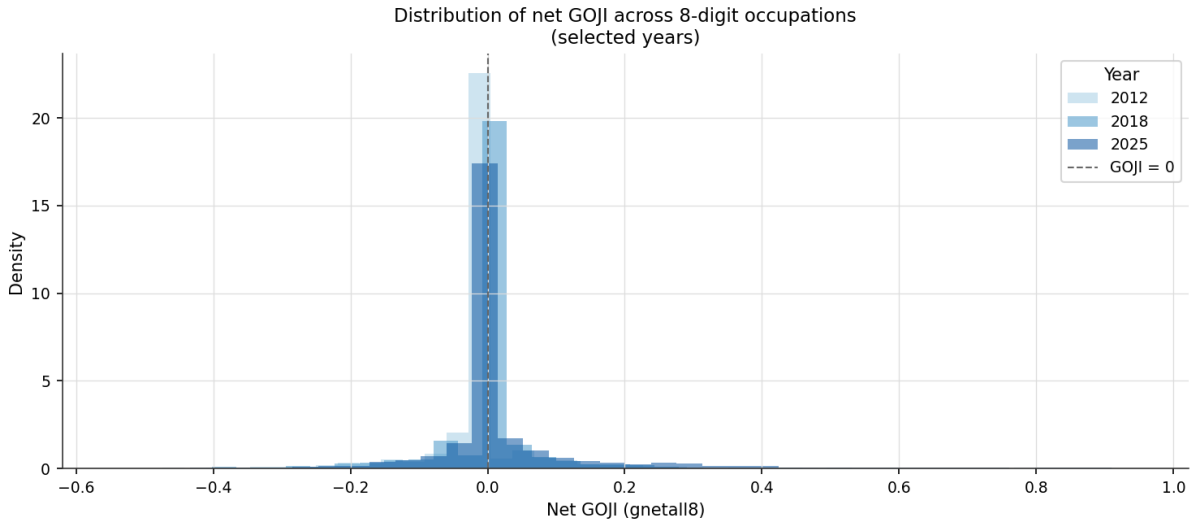


Figure 11: Kernel-density estimate of net GOJI across all 8-digit single occupations (DKZ), selected years 2012, 2018, and 2025. Vertical dashed line marks $GOJI^{\text{net}} = 0$ (purely white-task occupations). The distribution is zero-spike with thin green and brown tails. **Source:** GOJI 4.0 (v1225), own calculations from BERUFENET.

C.2 Employment-Weighted Greenness Group Distribution Over Time

Figure 12 shows the employment-weighted share of the five greenness groups (dark brown through dark green, defined by the ± 0.1 net-GOJI thresholds) for each year 2013–2025.

The dominant feature is the stability of the neutral (white) category at around 62–67 % of employment through 2022, followed by a sharp fall to roughly 50 % in 2024–2025 as the BA’s green-leaf rollout in BERUFENET substantially enlarged the light-green and dark-green pools. Brown-category employment shares decline gradually throughout the period, particularly for the light-brown group.

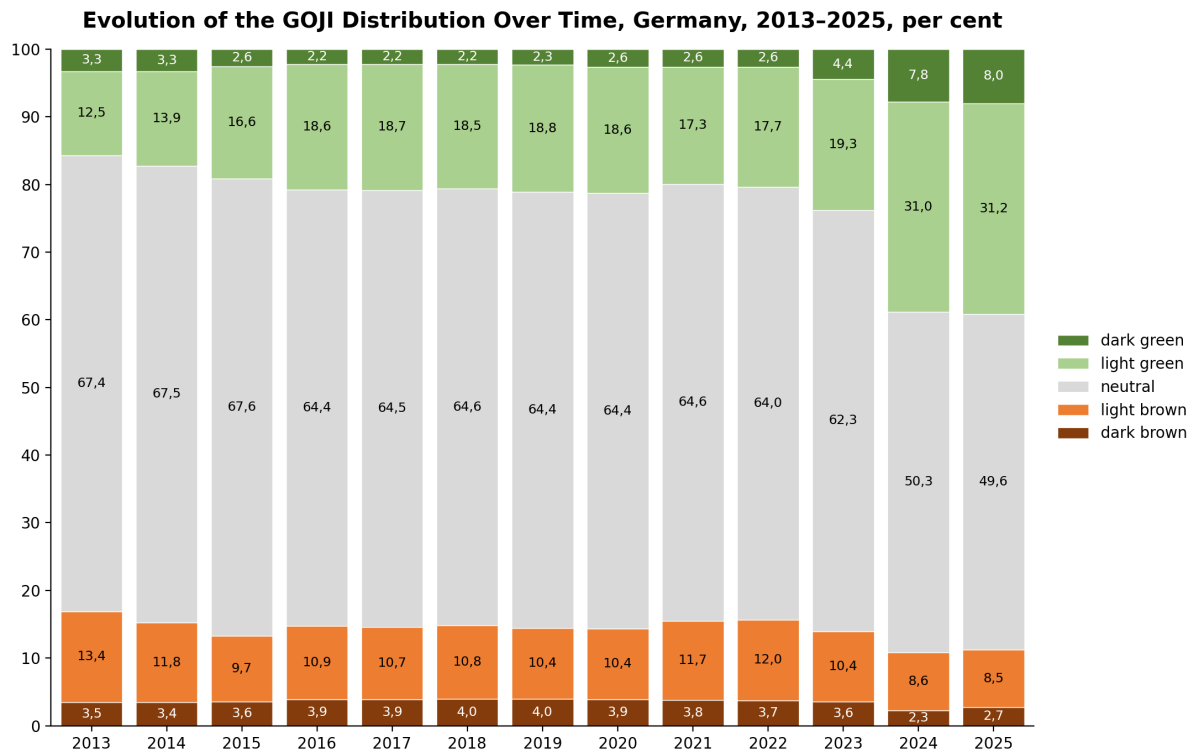


Figure 12: Employment-weighted share (%) of the five GOJI greenness groups (dark brown / light brown / neutral / light green / dark green) per year, Germany 2013–2025. Groups defined by net GOJI thresholds: dark brown < -0.1 , light brown $\in [-0.1, 0)$, neutral = 0, light green $\in (0, 0.1]$, dark green > 0.1 . **Data sources:** GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi, KldB2010 5-digit, from 2013).

D Appendix D: Availability of GOJI in the IAB Occupational Panel

The GOJI is integrated in the publicly accessible IAB Occupational Panel, which is released under the Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0) license.³ The most recent version (IAB Occupational Panel 2012–2022, referred to as `iab-occpan1222v2`) includes GOJI 3.0 (v1222) for the period 2012–2022, at the 3+5 level of the KldB 2010 classification (i.e.:. The documentation of the prior Panel version is provided by Grienberger, Janser–Lehmer (2022), while changes in the 2022 update—including the incorporation of GOJI and related metadata—are listed in the Panel’s change log (IAB 2025).⁴ In the upcoming version of the IAB Occupational Panel, covering the years 2012 to 2025, GOJI 4.0 (v1225) will also be included.

³The IAB Occupational Panel can be downloaded here: <https://iab.de/en/daten/iab-occupational-panel/>.

⁴https://doku.iab.de/arbeitsmarktdaten/ChangeLog_OccPan1222v2_EN_20250311.txt.

E Appendix E: Version Comparison — Five Generations of GOJI

The Greenness-of-Jobs Index was developed in successive versions from 2016 to 2025, each refining the task classification vocabulary and the aggregation methodology. This appendix provides a systematic, side-by-side comparison of all five versions to document methodological progress and to validate the current production version (v1225). The comparison is built on a *single, uniform* processing pipeline: occupation-level GOJI values are drawn directly from each version’s published 5-digit panel and reported separately for three metrics — GOJI^+ (green share), GOJI^- (brown share), and $\text{GOJI}^{\text{net}} = \text{GOJI}^+ - \text{GOJI}^-$ (net index). GOJI^- and GOJI^{net} are available from v1220 onward; v1216 is a green-only index with no brown component. Employment weights are used as stored, and the five greenness categories (dark brown to dark green, with “white” at exactly zero) are re-derived from the same net cut-offs (± 0.1) for every version that has a net index so that any observed difference reflects the versions themselves, not the analysis. All underlying files are produced by `prog/compare_versions.py`.

E.1 E.1 Version Overview

Table 8 summarises the five versions. The index evolved from a green-only count (v1216, produced for the 2016 PhD thesis) through a version with separate green and brown text-mining scores (v1220, used in REGIO.Trans and the KB 2023 report), to a unified net index built on the official BA Green-Skills list (v1222 onward). The green task vocabulary — the official *Kompetenzliste* of the Bundesagentur für Arbeit — was published on 24 November 2023 and has been the authoritative green source since v1222. Brown tasks were text-mined in v1220/1222 and manually curated in v1225. The unpublished intermediate version 1224 is excluded from this appendix; for a full five-version comparison and a detailed account of v1224’s imputation error, see the internal technical note `doc_orga/v1224vsv1225/`.

E.2 E.2 National Employment-Weighted GOJI (GOJI-D)

Tables 9–11 and Figure 13 report the national employment-weighted GOJI-D by version and year, separated by metric: GOJI^+ (green share, Table E2a), GOJI^- (brown share, Table E2b), and GOJI^{net} (green minus brown, Table E2c; not applicable for v1216). Three findings stand out. First, the *green share* (GOJI^+ , Table E2a) rises substantially across all versions in 2023–2024, driven by the BA green-task roll-out in BERUFENET (Section E.7). v1216 and v1220 show the highest green shares (+0.015–+0.019) because their vocabulary was larger; from v1222 onward the official BA list yields smaller but more authoritative green counts. Second, the *brown share* (GOJI^- , Table E2b) reveals the direct impact of the brown-list choice: v1220 (text-mined, ~ 290 – 310 patterns) has the largest brown share; v1222 is moderately high (~ 255 – 306 patterns); v1225 is the smallest (190 patterns after curation). Third, net-level differences (GOJI^{net} , Table E2c) follow directly from the green-minus-brown arithmetic. v1222 is *slightly negative* throughout (-0.003 to -0.000) because the strict BA green list cut the green count sharply while the text-mined brown list remained large, flipping the national net negative. v1225 returns to near-zero positive (+0.000–+0.020) because the curated brown list is smaller. These differences are therefore *expected* and interpretable as direct consequences of the respective classification choices.

E.3 E.3 Occupational Distribution across Greenness Categories

Figure 14 shows, for each version, the annual employment share that falls into each of the five greenness categories: dark brown (≤ -0.1), light brown ($(-0.1, 0)$), white ($= 0$), light green ($(0, 0.1)$), and dark green (≥ 0.1). The categories are derived from the net GOJI with the same ± 0.1 cut-offs for every version. The dominant class throughout is “white”, reflecting the large number of occupations with no classified green or brown tasks. The share of light-green occupations grows substantially in 2023–2024 across all consistent versions with a net index (v1220/1222/1225), driven by the BERUFENET roll-out. v1216 is absent from this figure because it has no net GOJI and therefore no net-based greenness categories.

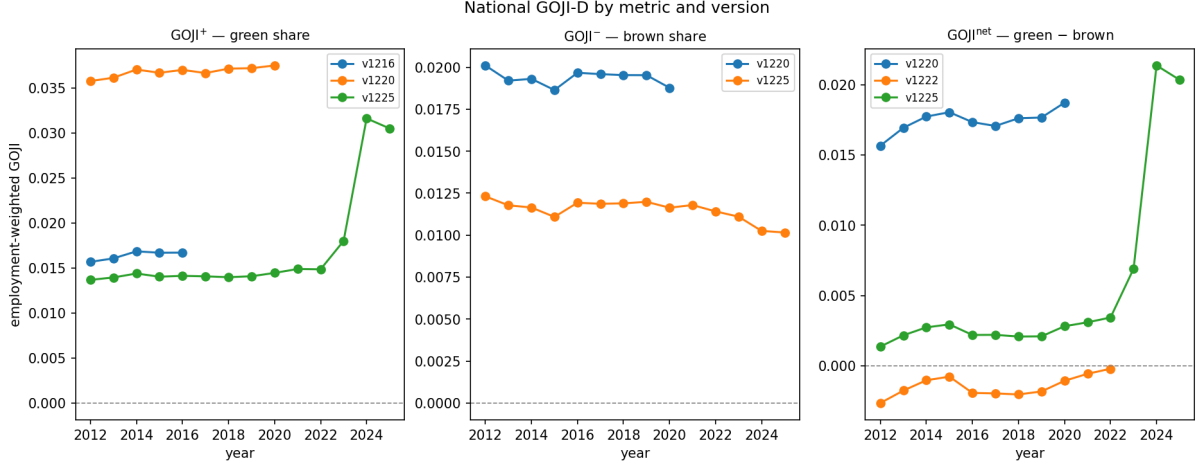


Figure 13: National employment-weighted GOJI-D by version and year (three metrics). *Left panel:* GOJI⁺ (green share) including v1216 (green-only index). *Centre panel:* GOJI⁻ (brown share); v1216 absent (no brown list). *Right panel:* GOJI^{net} = green — brown; v1216 absent. v1224 deviates markedly in 2013–2022 due to a wrong-aggregate imputation error (Section ??). Source: own calculations; data from BERUFENET / BA employment statistics.



Figure 14: Employment-share distribution across the five GOJI greenness categories by version (small multiples; v1216 excluded — no net GOJI). Categories re-derived uniformly from net GOJI with ± 0.1 cut-offs for each version. Categories: dark brown, light brown, white (= 0), light green, dark green. Source: own calculations.

E.4 E.4 Task Dictionary Evolution

Table 12 and Figure 15 track the number of distinct classified tasks in the green and brown lists for each version over time. Several patterns are noteworthy:

- **Green list:** v1216 and v1220 used internally developed text-mining vocabularies of 150–220 green task patterns; from v1222 onward the source is the official BA Green-Skills list, which yields a smaller but more authoritative set of around 130–

160 green tasks up to 2022. The large jump from 160 (2022) to 289 (2023) to 376 (2024) reflects the BERUFENET data roll-out described in Section E.7.

- **Brown list:** text-mining in v1220 produced roughly 290–310 brown patterns; manual curation narrowed this to 255–306 in v1222, and further to 163 (v1224) and 184–190 (v1225). The reduction reflects the removal of patterns that flagged too many occupations incorrectly.
- **Green-brown overlap:** v1220’s text-mining flagged 7–9 tasks simultaneously as green *and* brown in some years — a fuzzy text-mining artefact not present in later curated versions.

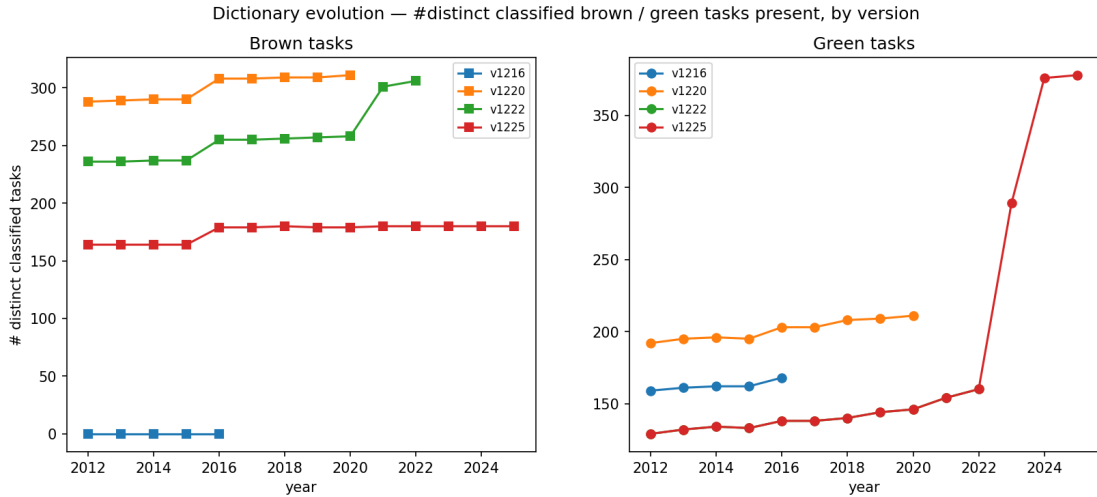


Figure 15: Number of distinct classified tasks in the brown list (left) and green list (right) by version and year. v1216 had no brown list. The large green ramp in 2023–2024 is driven by the official BA Green-Skills list roll-out in BERUFENET (Section E.7). Source: own calculations from version-specific classification files.

E.5 E.5 Classified Requirements-Catalog Composition

Figure 16 decomposes the full classified requirements catalog into its three colour classes — brown, white, and green — for each version over time. White tasks are by far the dominant class, comprising over 90% of all classified tasks in every year and version. The green share rises sharply in 2023–2024 in line with the dictionary expansion. The total

catalog size grows from roughly 6,500 tasks in 2012 to over 8,700 in 2025, reflecting new occupations and task descriptions added by the BA over the period.

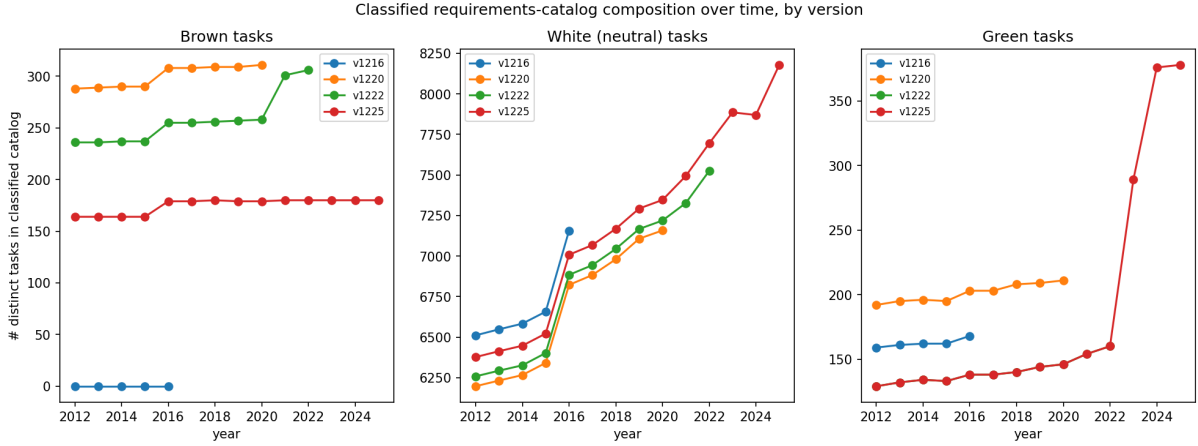


Figure 16: Number of distinct tasks in each colour class of the classified requirements catalog: brown (left), white (centre), green (right), by version and year. White tasks dominate throughout. Source: own calculations.

E.6 E.6 Example Occupations

Tables 13–15 and Figure 17 track three occupations chosen to illustrate the full range of GOJI profiles across all three metrics (green share, brown share, and net): a robustly green occupation (roofer, KldB 32142), a robustly brown one (construction materials production specialist, KldB 21222), and an occupation transitioning from white (GOJI = 0) to light green over time (mechatronics specialist, KldB 26112).

- **Roofer (Dachdeckerei – Fachkraft, 32142):** Green in every version and every year due to tasks involving photovoltaics, solar-thermal systems, green roofs, and building insulation. The exception is v1224’s *panel* for 2013–2023, which incorrectly reports negative GOJI values — even though the v1224 task matrix classifies the roofer as green throughout. This discrepancy is the key diagnostic signal for the 1224 imputation error (Section ??).
- **Construction materials production (21222):** Brown in every version that measures a net index (GOJI around -0.14 to -0.30 across versions and years), reflecting energy-intensive production processes. v1216 appears slightly positive

(+0.011) because it is a green-only index with no brown subtraction; the occupation does carry a small number of green tasks (e.g. energy-efficiency advisory duties). The true net colour is unambiguously brown once brown tasks are counted.

- **Mechatronics (26112):** Net GOJI near zero through 2022; rises to light green from 2024 onward as BERUFENET assigns new green tasks (e-mobility, energy efficiency maintenance) to this occupation.

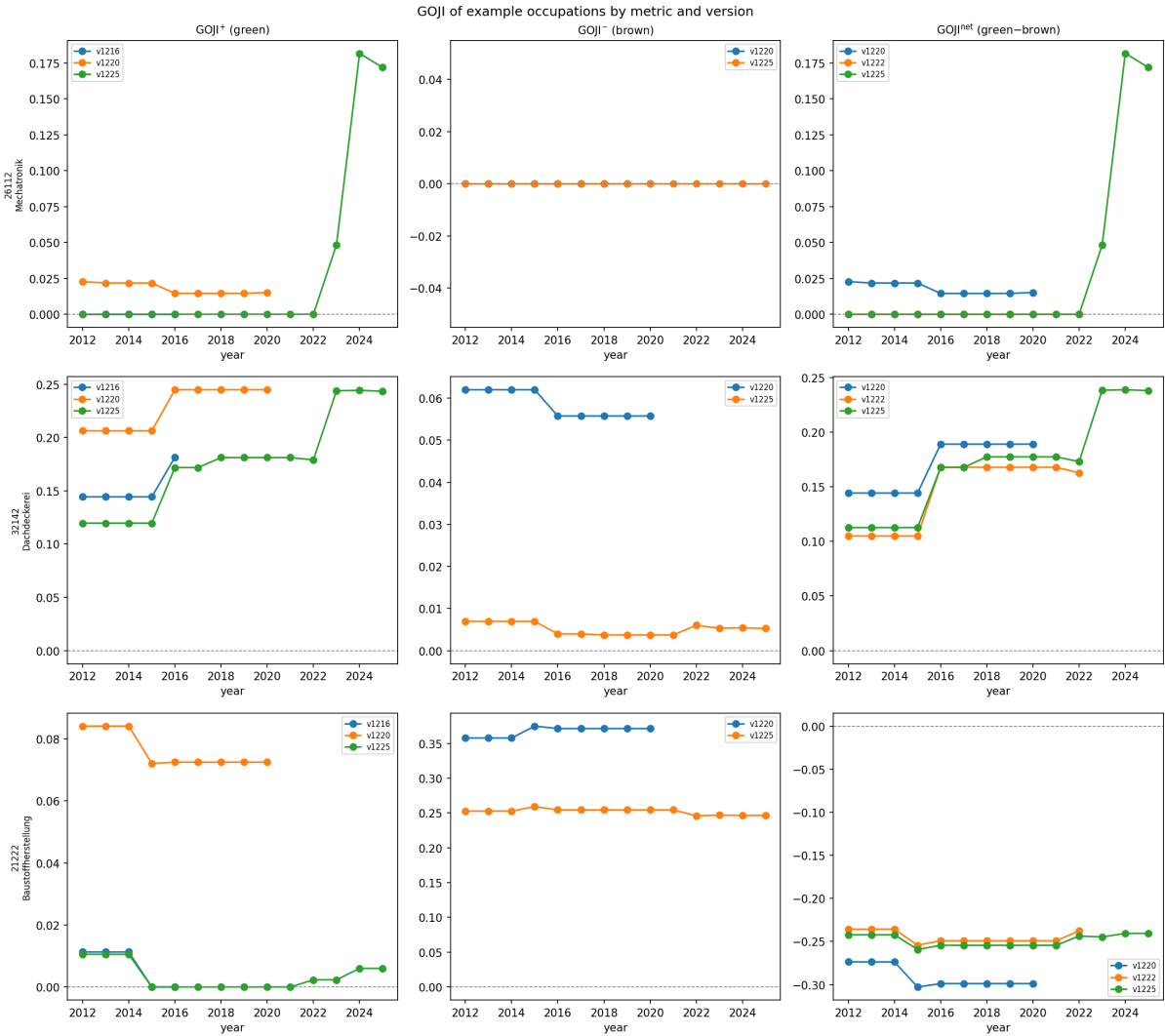


Figure 17: GOJI of three example occupations over time (3×3 grid: rows = occupations, columns = metric). *Left column:* GOJI⁺ (green share); v1216 included. *Centre column:* GOJI⁻ (brown share); v1216 absent (no brown list). *Right column:* GOJI^{net}; v1216 absent. Source: own calculations; BERUFENET.

E.7 E.7 Timing of the Green Roll-out in BERUFENET

The sharp rise in green-classified tasks visible in 2023 prompts a question: if the BA Green-Skills list was published on 24 November 2023, why does the index rise *in* 2023 and not only in 2024?

The answer lies in the snapshot dates and in the distinction between the task *catalog* and the occupation–task *assignment matrix*. The GOJI green count is driven by green-listed tasks *present in the data each year*; the green dictionary itself is applied uniformly across all years, so year-to-year movement reflects the BERUFENET data, not changes to the list. Two layers explain the apparent 2023 jump. First, the snapshots are dated: the 2022 requirements matrix is from October 2022 (before the green list was published), whereas the 2023 matrix is from **December 2023** — one month *after* publication. By that December-2023 snapshot the BA had already entered almost the entire green vocabulary into the requirements *catalog* (368 of 378 green tasks present, versus 166 in 2022). Second, attaching those green tasks to individual occupation *profiles* (the occupation×task assignment matrix) lagged: only 289 of the catalogued green tasks were assigned to at least one occupation in the 2023 snapshot, rising to 376 by the October-2024 snapshot. Because a task only raises an occupation’s GOJI once it is *assigned to that occupation*, the index rises in two steps — a large first wave in the year-end-2023 snapshot and the remainder in 2024 — even though the green-skills list was a single end-of-2023 event. The apparent “2023” timing is therefore really “December 2023,” and the green skills were not “only implemented in 2024”: the vocabulary arrived at end-2023 and the occupation-level assignment finished in 2024.

(Source: *catalog vs. matrix green-task counts per year, this analysis; BERUFENET snapshot dates: 31 Oct 2022, 19 Dec 2023, 18 Oct 2024; BA Green-Skills list dated 24 Nov 2023.*)

F Appendix F: Previous Research on Green Jobs and Occupational Greenness

F.1 F.1 Defining and Measuring Green Jobs

One of the earliest challenges was agreeing on what constitutes a “green job.” The debate revolved largely around whether to use an output-based or process-based definition. Early 2000s work such as [Rennings and Zwick \(2002\)](#) and policy initiatives by the ILO and UNEP laid conceptual groundwork, focusing on employment in environmental sectors. However, as [Deschenes \(2013\)](#) notes, estimates of green employment vary widely depending on definitions and measurement approaches. The U.S. BLS Green Jobs Initiative (2010–2013) identified green jobs by both an output approach (employment in firms producing environmental goods and services) and a process approach (employment in firms using environmentally friendly practices). Results summarised by [Sommers \(2013\)](#) indicated that the green goods and services sector accounted for roughly 2–3 % of total U.S. employment, but significant additional green-process employment existed in conventional industries — a point also underscored by [Peters \(2014\)](#), who identified 176 occupations (roughly 20 % of the U.S. occupational classification) with material green task content. European statistical agencies estimated the Environmental Goods and Services Sector (EGSS) along similar lines; [Horbach and Janser \(2016\)](#) further found that German establishments in green markets exhibited above-average employment growth in 2006–2012.

The limitations of broad output- and process-based definitions prompted growing interest in task-based measurement. [Dierdorff et al. \(2009\)](#) were among the first to classify occupations by how the green economy affects them — distinguishing Green Increased Demand occupations, Green Enhanced Skills occupations, and Green New and Emerging occupations using the O*NET database. Their finding that few genuinely new occupations emerge (most green change reflects changing task content within existing titles) motivates the within-occupation, dynamic perspective of GOJI. [Consoli et al. \(2016\)](#) compared green and non-green occupations using O*NET and found that green occupa-

tions tend to require above-average formal education and interpersonal skills. [Vona et al. \(2018\)](#) examined skill gaps between green and brown occupations and concluded that the gap is modest — green occupations require an additional dose of specific technical knowledge, but the general skill base is largely shared with comparable brown occupations, implying that re-skilling barriers may be lower than feared.

[Vona et al. \(2019\)](#) further documented that green employment in the U.S. is concentrated among higher-skilled, higher-wage, and predominantly male workers, raising concerns about gender equity and regional inequality in the green transition. The OECD working paper by [Tyros et al. \(2023\)](#) argues that most workers in polluting occupations hold skills compatible with greener roles except those in highly automatable positions. Online vacancy data offer a complementary perspective: [Sato et al. \(2023\)](#) find that both low-carbon and high-carbon job postings list above-average skill requirements, with a smaller gap between the two than between either and generic jobs, suggesting that worker transition paths may be shorter than commonly assumed.

A recent critique by Villani, González Vázquez, and Fernández-Macías (2025) argues that task-based green-employment indicators have not shown statistically significant growth across 24 European countries and correlate weakly with actual environmental outcomes. This points to measurement limitations inherent in static occupational classifications applied to dynamic labour markets. The GOJI framework partly mitigates this concern through annual BERUFENET updates and the use of official BA task identifiers rather than keyword proximity — but the caution about interpreting GOJI values as measures of actual ecological impact rather than potential task content is noted in the main text (Section 3.1).

F.2 F.2 Labor Market Impacts of the Green Transition

For the full literature review on labor market impacts (employment effects of climate policy, sectoral reallocation, transition risks), see `doc_paper/LaTeX/_temp/literature_labor_market_imp`

Table 1: Employment-weighted GOJI by Occupational Segment and Requirements Level, 2013 and 2025 (GOJI 4.0, v1225)

Segment $\times$ ReqLev	Net GOJI			Green share (GOJI ⁺)			Brown share (GOJI ⁻)		
	2013	2025	Δ	2013	2025	Δ	2013	2025	Δ
S11.1	-0.001	0.002	+0.004	0.022	0.023	+0.000	0.024	0.020	-0.003
S11.2	0.105	0.180	+0.075	0.127	0.200	+0.073	0.022	0.019	-0.002
S11.3	0.131	0.251	+0.121	0.160	0.273	+0.113	0.029	0.022	-0.008
S11.4	0.112	0.307	+0.195	0.146	0.319	+0.173	0.033	0.012	-0.021
S12.1	-0.037	-0.041	-0.004	0.000	0.000	+0.000	0.037	0.041	+0.004
S12.2	-0.055	-0.041	+0.014	0.002	0.019	+0.017	0.057	0.060	+0.003
S12.3	-0.025	-0.008	+0.017	0.002	0.012	+0.010	0.027	0.020	-0.007
S12.4	-0.015	0.000	+0.016	0.011	0.018	+0.007	0.026	0.017	-0.009
S13.1	-0.038	-0.035	+0.004	0.016	0.019	+0.003	0.055	0.053	-0.001
S13.2	0.015	0.067	+0.052	0.024	0.081	+0.057	0.009	0.015	+0.005
S13.3	0.010	0.058	+0.048	0.015	0.062	+0.047	0.004	0.004	-0.001
S13.4	0.009	0.055	+0.046	0.013	0.058	+0.046	0.004	0.003	-0.000
S14.1	-0.042	-0.043	-0.000	0.078	0.070	-0.008	0.121	0.113	-0.008
S14.2	0.014	0.081	+0.067	0.061	0.127	+0.066	0.047	0.046	-0.001
S14.3	-0.008	0.134	+0.142	0.036	0.159	+0.123	0.044	0.024	-0.019
S14.4	-0.020	0.212	+0.232	0.026	0.242	+0.215	0.047	0.030	-0.017
S21.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S21.2	-0.013	-0.006	+0.007	0.002	0.012	+0.010	0.015	0.018	+0.003
S21.3	-0.009	0.005	+0.014	0.001	0.010	+0.009	0.010	0.005	-0.004
S21.4	-0.001	0.002	+0.003	0.000	0.004	+0.004	0.001	0.002	+0.001
S22.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S22.2	0.000	0.003	+0.003	0.000	0.003	+0.003	0.000	0.000	+0.000
S22.3	-0.001	-0.001	+0.000	0.001	0.001	+0.001	0.002	0.002	+0.000
S22.4	0.002	0.005	+0.003	0.002	0.005	+0.003	0.000	0.000	+0.000
S23.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S23.2	-0.000	0.020	+0.020	0.000	0.020	+0.020	0.000	0.000	-0.000
S23.3	0.006	0.005	-0.001	0.007	0.005	-0.002	0.001	0.000	-0.001
S23.4	0.000	0.004	+0.004	0.001	0.004	+0.004	0.000	0.000	-0.000
S31.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S31.2	0.001	0.015	+0.014	0.002	0.017	+0.015	0.001	0.002	+0.001
S31.3	0.001	0.004	+0.003	0.001	0.004	+0.003	0.000	0.000	+0.000
S31.4	-0.001	0.007	+0.008	0.000	0.009	+0.009	0.001	0.002	+0.001
S32.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S32.2	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S32.3	-0.000	0.000	+0.001	0.000	0.001	+0.000	0.000	0.000	-0.000
S32.4	0.001	0.004	+0.004	0.001	0.004	+0.004	0.000	0.000	+0.000
S33.1	0.000	0.000	+0.000	0.000	0.000	+0.000	0.000	0.000	+0.000
S33.2	0.000	0.016	+0.016	0.000	0.016	+0.016	0.000	0.000	+0.000
S33.3	0.000	0.012	+0.012	0.000	0.012	+0.012	0.000	0.000	+0.000
S33.4	0.002	0.009	+0.007	0.002	0.011	+0.008	0.000	0.001	+0.001
S41.1	-0.053	-0.059	-0.006	0.000	0.000	+0.000	0.053	0.059	+0.006
S41.2	0.008	0.020	+0.012	0.035	0.041	+0.006	0.027	0.021	-0.006
S41.3	0.011	0.041	+0.030	0.012	0.041	+0.029	0.001	0.000	-0.001
S41.4	0.017	0.047	+0.030	0.021	0.048	+0.027	0.004	0.001	-0.003
S51.1	0.000	-0.012	-0.012	0.000	0.000	+0.000	0.000	0.012	+0.012
S51.2	0.012	0.010	-0.002	0.026	0.018	-0.008	0.014	0.008	-0.006
S51.3	0.055	0.055	+0.000	0.055	0.055	+0.000	0.000	0.000	+0.000
S51.4	0.073	0.027	-0.046	0.073	0.027	-0.046	0.000	0.000	+0.000
S52.1	0.002	0.011	+0.009	0.002	0.011	+0.009	0.000	0.000	+0.000
S52.2	0.017	0.031	+0.014	0.061	0.065	+0.003	0.044	0.034	-0.011
S52.3	-0.006	0.015	+0.021	0.027	0.039	+0.012	0.034	0.024	-0.009

Table 4: Employment-weighted GOJI by WZ08 One-Digit Section, 2013 and 2025 (GOJI 4.0, v1225)

Sec.	Industry	Net GOJI			Green (GOJI ⁺)			Brown (GOJI ⁻)	
		2013	2025	Δ	2013	2025	Δ	2013	2025
A	Agriculture, forestry, fishing	0.018	0.059	+0.040	0.059	0.095	+0.036	0.041	0.03
B	Mining and quarrying	-0.024	0.006	+0.030	0.021	0.046	+0.025	0.044	0.03
C	Manufacturing	-0.012	0.006	+0.019	0.009	0.029	+0.019	0.022	0.02
D	Electricity, gas, steam	0.027	0.077	+0.050	0.032	0.082	+0.050	0.005	0.00
E	Water supply, waste management	0.082	0.115	+0.033	0.110	0.139	+0.029	0.028	0.02
F	Construction	-0.017	0.025	+0.041	0.031	0.074	+0.043	0.048	0.04
G	Wholesale and retail trade	0.000	0.017	+0.017	0.007	0.023	+0.016	0.006	0.00
H	Transportation and storage	0.017	0.038	+0.021	0.052	0.064	+0.012	0.035	0.02
I	Accommodation and food service	-0.001	0.007	+0.008	0.004	0.011	+0.007	0.004	0.00
J	Information and communication	0.001	0.014	+0.012	0.002	0.014	+0.012	0.001	0.00
K	Financial and insurance activities	0.001	0.016	+0.015	0.001	0.016	+0.015	0.000	0.00
L	Real estate activities	0.014	0.041	+0.028	0.019	0.045	+0.027	0.005	0.00
M	Professional, scientific, technical	0.003	0.031	+0.028	0.009	0.036	+0.027	0.006	0.00
N	Administrative and support services	0.016	0.029	+0.013	0.027	0.037	+0.010	0.011	0.00
O	Public administration and defence	0.011	0.038	+0.026	0.017	0.042	+0.025	0.005	0.00
P	Education	0.004	0.014	+0.011	0.006	0.015	+0.010	0.002	0.00
Q	Human health and social work	0.002	0.004	+0.002	0.003	0.006	+0.003	0.001	0.00
R	Arts, entertainment and recreation	0.006	0.012	+0.006	0.009	0.015	+0.007	0.003	0.00
S	Other service activities	0.007	0.019	+0.013	0.010	0.022	+0.012	0.003	0.00
T	Households as employers	0.005	0.009	+0.004	0.007	0.011	+0.004	0.002	0.00
U	Extraterritorial organisations	0.003	0.020	+0.017	0.014	0.029	+0.015	0.011	0.00

Data sources: GOJI 4.0 (v1225), BA employment statistics (BST/SVB-ohne-Azubi, KldB2010 $\times$ WZ08 3-digit).

Table 5: Occupation Vignette: Ingenieur/in - Bau (KldB 31104115, 2025 BERUFENET snapshot, GOJI 4.0)

Task (German requirement identifier / translated)	Category
Baubiologie	Green
Drei-Liter-Haus-Standard	Green
Energetische Gebäudedämmung	Green
Energetische Stadtsanierung	Green
Erosionsschutz	Green
Gebäudeenergiegesetz (GEG)	Green
Nachhaltige Baustoffe	Green
Nachhaltige Dämmstoffe	Green
Nullenergiehaus-Standard	Green
Passivhaus-Standard	Green
Plusenergiehaus-Standard	Green
Umwelt-Produktdeklaration Bauprodukte	Green
Urban Mining	Green
Ökologisches Bauen	Green
Betonbau	Brown
Betontechnologie	Brown
Stahlbaukonstruktion (Planung, Überwachung)	Brown
Stahlbetonbau	Brown
Apps für Planung und Aufmaß	White
Ausschreibung, Vergabe, Leistungsbeschreibung	White
Barrierefreies Bauen	White
Bau- und Architektenrecht	White
Bauabnahme	White
Baubetrieb	White
Baukonstruktion	White
Bauleitung	White
Bauphysik	White
Baustatik	White
Baustoffkunde, -technologie	White
Baustoffprüfung	White
Bauwerksprüfungen	White
Betonsanierung	White
Bodenmechanik	White
Brückenbau	White
Building Information Modeling - BIM	White
CAD (Computer Aided Design)-Systeme anwenden	White
Dimensionierung	White
Freiraumplanung	White
Grundriss (erstellen, bearbeiten etc.)	White
Industrie-, Gewerbebau (Planung, Überwachung)	White
KI-gestützte Stadtentwicklung	White
KI-gestütztes Generatives Design	White
Konstruktiver Ingenieurbau	White
Messmethoden (Umgang, bestimmen etc.)	White
Revisionspläne (erstellen etc.)	White
Straßenbau	White
Tunnelbau	White
Umbaumaßnahmen (organisieren, kontrollieren etc.)	White
Vermessung	White
Wohnungsbau (Planung, Überwachung)	White

Total tasks: 50; green: 14 (GOJI⁺ = 0.280); brown:

Table 6: Development Phases of the GOJI Index

Version	Years 2012 to	Key advance	Green source	Brown source
GOJI 1.0 (v1216)	2016	First task-based green index	Text mining	—
GOJI 2.0 (v1220)	2020	Added brown/white/net; groups 3e & 5e	Text mining	Text mining
GOJI 3.0 (v1222)	2022	Official BA Green Skills list; imputa- tion	BA list (TM as QC)	Curated dict. (TM)
GOJI 4.0 (v1225)	2025	Updated lists; Python; BERUFENET green-leaf rollout visible	BA list, up- dated (TM as QC)	Curated dict., 190 tasks (TM)

Table 7: Comparison of GOJI Methodologies across Four Published Versions

Aspect	GOJI 1.0 (v1216)	GOJI 2.0 (v1220)	GOJI 3.0 (v1222)	GOJI 4.0 (v1225)
Task categories	Green only	Green, Brown, White	Green, Brown, White	Green, Brown, White
Index compo- nents	<i>GOJI</i> ⁺ (green share)	<i>GOJI</i> ⁺ , <i>GOJI</i> ⁻ , net	<i>GOJI</i> ⁺ , <i>GOJI</i> ⁻ , net	<i>GOJI</i> ⁺ , <i>GOJI</i> ⁻ , net
Green source	Text mining (~300 key- words)	Text mining	Official BA Green Skills list (TM as QC)	BA list, up- dated 378 tasks (TM as QC)
Brown source	Not used	Text mining (~100 key- words)	Curated dict. (TM)	Curated dict., 190 tasks (TM)
Imputation	No	No	Yes (new)	Yes
Data years	2012–2016	2012–2020	2012–2022	2012–2025
Pipeline	Stata	Stata	Stata	Python (new)

Table 8: Overview of the four GOJI versions covered in this document. “Used for” refers to the publication in which the version first appeared. GOJI 1224 is excluded from this document (unpublished; imputation error).

Version	Years	Type	Green source	Brown source	Imputed	Used for
v1216	2012–2016	green only	text mining + curated	—	no	PhD th
v1220	2012–2020	green & brown (separate)	text mining + curated	text mining + curated	no	REGIO
v1222	2012–2022	net	official green list	BA text mining + curated	yes	GOJI f
v1225	2012–2025	net	official green list	BA curated (211)	yes	no pub

Table 9: National employment-weighted GOJI⁺ (green share) by version and year (v1216, v1220, v1222, v1225). v1216 is a green-only index; all other versions’ green shares are the gross positive component before netting out brown. Empty cells: years not covered by that version.

Year	v1216	v1220	v1225
2012	+0.0157	+0.0358	+0.0137
2013	+0.0161	+0.0361	+0.0139
2014	+0.0168	+0.0370	+0.0144
2015	+0.0167	+0.0367	+0.0140
2016	+0.0167	+0.0370	+0.0141
2017		+0.0366	+0.0141
2018		+0.0371	+0.0140
2019		+0.0372	+0.0141
2020		+0.0375	+0.0144
2021			+0.0149
2022			+0.0149
2023			+0.0180
2024			+0.0316
2025			+0.0305

Table 10: National employment-weighted GOJI^- (brown share) by version and year (v1220, v1222, v1225). v1216 had no brown component and is absent. Empty cells: years not covered by that version.

Year	v1220	v1225
2012	+0.0201	+0.0123
2013	+0.0192	+0.0118
2014	+0.0193	+0.0116
2015	+0.0186	+0.0111
2016	+0.0197	+0.0119
2017	+0.0196	+0.0119
2018	+0.0195	+0.0119
2019	+0.0195	+0.0120
2020	+0.0188	+0.0116
2021		+0.0118
2022		+0.0114
2023		+0.0111
2024		+0.0103
2025		+0.0102

Table 11: National employment-weighted GOJI^{net} (green – brown) by version and year (v1220, v1222, v1225). v1216 is absent (no net concept for a green-only index). Empty cells: years not covered by that version.

Year	v1220	v1222	v1225
2012	+0.0157	-0.0026	+0.0014
2013	+0.0169	-0.0018	+0.0022
2014	+0.0177	-0.0010	+0.0027
2015	+0.0180	-0.0008	+0.0029
2016	+0.0173	-0.0019	+0.0022
2017	+0.0171	-0.0020	+0.0022
2018	+0.0176	-0.0020	+0.0021
2019	+0.0177	-0.0018	+0.0021
2020	+0.0187	-0.0011	+0.0028
2021		-0.0006	+0.0031
2022		-0.0002	+0.0034
2023			+0.0069
2024			+0.0214
2025			+0.0203

Table 12: Number of distinct classified tasks in the green and brown lists per GOJI version and year (“Grn” = green, “Brn” = brown). GOJI 1216 has no brown list (green-only index); empty cells mark years outside a version’s coverage. From v1222 onward, the official BA Green-Skills list (published 24 November 2023) is the source for green tasks.

Year	Grn v1216	Grn v1220	Grn v1222	Grn v1225	Brn v1216	Brn v1220	Brn v1222	Brn v1225
2012	159	192	129	129	0	288	236	164
2013	161	195	132	132	0	289	236	164
2014	162	196	134	134	0	290	237	164
2015	162	195	133	133	0	290	237	164
2016	168	203	138	138	0	308	255	179
2017		203	138	138		308	255	179
2018		208	140	140		309	256	180
2019		209	144	144		309	257	179
2020		211	146	146		311	258	179
2021			154	154			301	180
2022			160	160			306	180
2023				289				180
2024				376				180
2025				378				180

Table 13: GOJI⁺ (green share) of the three example occupations by version (v1216, v1220, v1222, v1225; selected years). All four versions report a positive green share for the roofer throughout.

Occupation (KldB)	Year	v1216	v1220	v1225
26112 — Mechatronik – Fachkraft				
	2012	+0.0000	+0.0227	+0.0000
	2016	+0.0000	+0.0145	+0.0000
	2020		+0.0152	+0.0000
	2022			+0.0000
	2024			+0.1817
	2025			+0.1719
32142 — Dachdeckerei – Fachkraft (roofer)				
	2012	+0.1443	+0.2063	+0.1196
	2016	+0.1815	+0.2448	+0.1717
	2020		+0.2448	+0.1811
	2022			+0.1790
	2024			+0.2443
	2025			+0.2433
21222 — Baustoffherstellung – Fachkraft				
	2012	+0.0113	+0.0841	+0.0105
	2016	+0.0000	+0.0725	+0.0000
	2020		+0.0725	+0.0000
	2022			+0.0023
	2024			+0.0060
	2025			+0.0060

Table 14: GOJI⁻ (brown share) of the three example occupations by version (v1220, v1222, v1225; selected years). v1216 is absent (no brown component). Construction materials (21222) has a high brown share throughout.

Occupation (KldB)	Year	v1220	v1225
26112 — Mechatronik – Fachkraft			
	2012	+0.0000	+0.0000
	2016	+0.0000	+0.0000
	2020	+0.0000	+0.0000
	2022		+0.0000
	2024		+0.0000
	2025		+0.0000
32142 — Dachdeckerei – Fachkraft (roofer)			
	2012	+0.0620	+0.0069
	2016	+0.0558	+0.0040
	2020	+0.0558	+0.0037
	2022		+0.0060
	2024		+0.0054
	2025		+0.0053
21222 — Baustoffherstellung – Fachkraft			
	2012	+0.3578	+0.2527
	2016	+0.3713	+0.2544
	2020	+0.3713	+0.2544
	2022		+0.2460
	2024		+0.2466
	2025		+0.2466

Table 15: GOJI^{net} (green – brown) of the three example occupations (v1220, v1222, v1225; selected years). v1216 absent.

Occupation (KldB)	Year	v1220	v1222	v1225
26112 — Mechatronik – Fachkraft				
	2012	+0.0227	+0.0000	+0.0000
	2016	+0.0145	+0.0000	+0.0000
	2020	+0.0152	+0.0000	+0.0000
	2022		+0.0000	+0.0000
	2024			+0.1817
	2025			+0.1719
32142 — Dachdeckerei – Fachkraft (roofer)				
	2012	+0.1443	+0.1050	+0.1127
	2016	+0.1890	+0.1679	+0.1678
	2020	+0.1890	+0.1679	+0.1774
	2022		+0.1628	+0.1730
	2024			+0.2389
	2025			+0.2381
21222 — Baustoffherstellung – Fachkraft				
	2012	-0.2736	-0.2357	-0.2422
	2016	-0.2989	-0.2492	-0.2544
	2020	-0.2989	-0.2492	-0.2544
	2022		-0.2377	-0.2437
	2024			-0.2406
	2025			-0.2406